

The world of plastic waste: A review

P.G.C. Nayanathara Thathsarani Pilapitiya, Amila Sandaruwan Ratnayake^{*,1}

Faculty of Applied Sciences, Uva Wellasa University, Passara Road, Badulla 90000, Sri Lanka

ARTICLE INFO

Keywords:

Plastic waste management
Environmental impacts
Marine pollution
Recycling
Circular economy

ABSTRACT

People discover various materials from time to time that break the boundaries of traditional materials. Plastic is a revolutionized material, and is referred to as “a material with 1,000 uses”. This review summarized up-to-date research on plastic and its waste pollution. Plastic has a domain throughout human life with its versatile properties such as lightweight, high durability, flexibility, and low production cost. This article describes the applications, benefits, production, consumption, and classifications of plastics. Plastic commercialization began with the Second World War and grew all over the world within less than a century. The global annual production of plastic is more than 359 million tons. Despite all the benefits, plastics cause severe environmental and public health issues. Accordingly, this study addresses the major issues of plastic waste on the environment and human health. Plastics can degrade into micro to nano sizes, and those fine particles are more spreadable in air, water, and soil. Therefore, both terrestrial and aquatic animals go through various negative impacts such as ingestion, entangling, ulcers, low reproduction, and oxidative stress. Microplastics also degrade human health due to cardiovascular diseases, chronic kidney disease, birth defects, cancer, etc. The closing contains the developed end-of-life options (e.g., recycling and reprocessing, incineration with energy recovery, modification reuse, value addition, and landfilling) of biodegradable and non-biodegradable plastic wastes. Several international, regional/national level legislations and policies/concepts (e.g., plastic trade, 3R policy, and circular economy) are available to manage plastic and plastic waste generation. Plastic waste management is also discussed offering practical insights and real-world scenarios. Solutions and challenges in effective plastic waste management guide to create a more sustainable and environmentally responsible approach. Finally, this review article highlights the importance of judicious decisions and the involvement of all stakeholders to overcome the plastic waste crisis.

1. Introduction

The world has witnessed significant changes through past centuries due to incredible inventions such as plastics. The statement “plastic is the greatest finding in the millennium”, is undeniable. According to the International Union of Pure and Applied Chemistry, plastic is a “polymeric material that may contain other substances to improve performance and/or reduce costs”. The first plastic was invented in 1860 and the beginning of the global plastic industry was marked in 1907. However, industrial development happened in the 1920s, and plastic production expanded in the 1940s (Fig. 1). In 1950, plastic production was around 2 million tons, and it was boosted to 368 million tons in 2019 (Fig. 1). Plastic consumption has thus increased about 180 times from 1950 to 2018. The worldwide plastic production is calculated to be 400.3 million tons in 2022. Plastic production is thus expected to increase exponentially hereafter (Rafey and Siddiqui, 2021). Organic

compounds such as synthetic or semi-synthetic substances (petrochemical, partially natural substances) have been used to form soft and flexible plastics. Typically, plastics can be categorized as ‘bio-based/organic polymers’ or ‘engineered’. Engineered plastics are developed using unrefined petroleum, flammable gas, and coal. In addition, about 4 % of fossil fuel is used to produce plastics. Sustainable raw materials such as carbs, vegetable fat/oils, microorganisms, and other natural substances are also used to produce bio-based plastics (Verma et al., 2016; Alhazmi et al., 2021).

Urbanization, economic development, and increasing population are the main drivers for the growth of plastic waste (Amasuomo and Baird, 2016). Waste can be generally declared as useless by-products/unwanted substances that are generated mainly during human activities and cause egregious results for both the environment and humans (Amasuomo and Baird, 2016; Hossain et al., 2022a,b). Meanwhile, the United Nations (UN) defined waste as “Materials that are not prime

* Corresponding author.

E-mail address: as_ratnayake@uwu.ac.lk (A.S. Ratnayake).

¹ <https://orcid.org/0000-0001-7871-2401>.

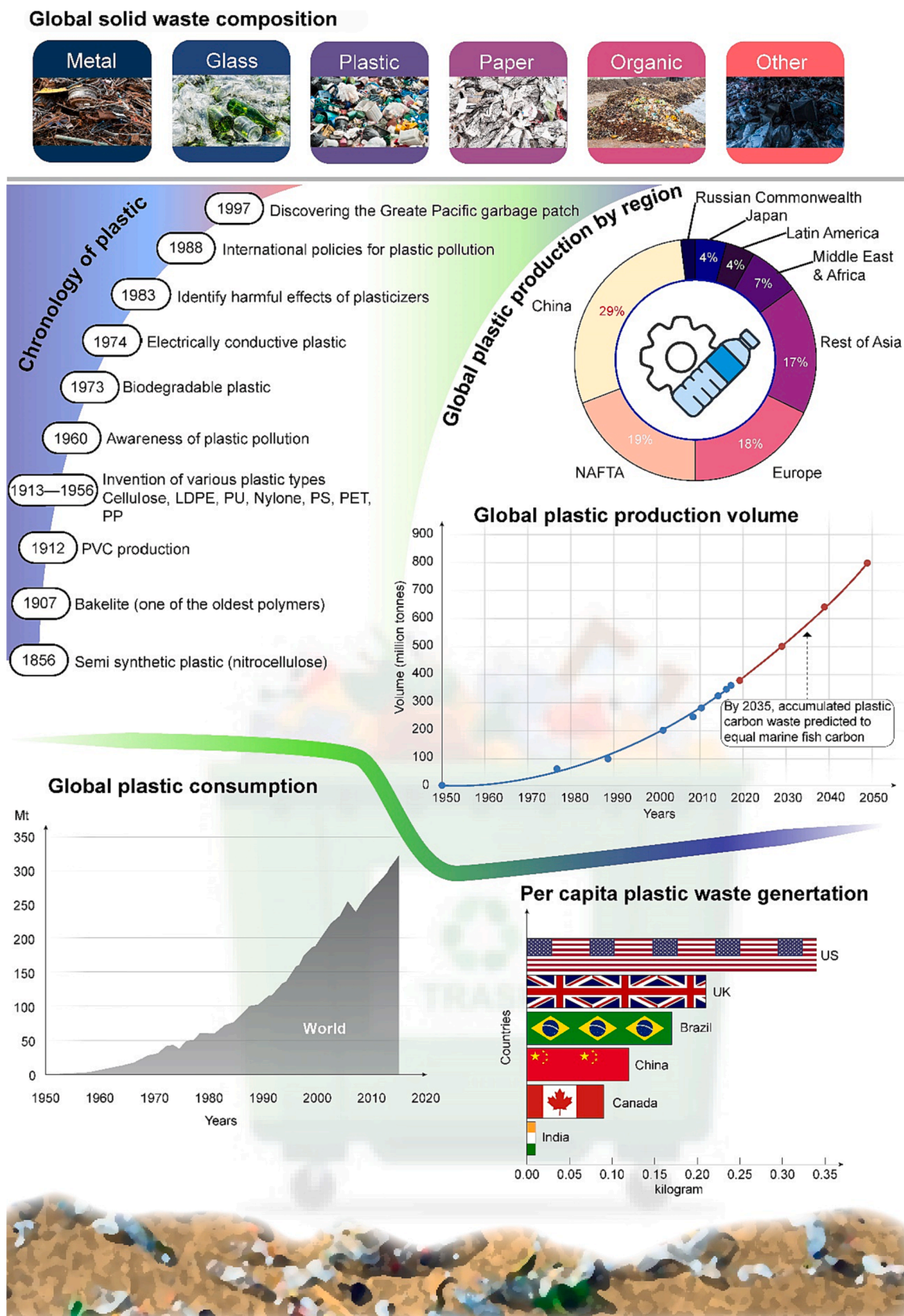


Fig. 1. Graphical representation of solid waste composition, important chronological events, plastic production by region, global plastic production and consumption, and per capita plastic waste generation.

products for which the generator has no further use in terms of his/her own purposes of production, transformation or consumption, and of which he/she wants to dispose of'. Even though waste generation has a long history, real problems are exposed after losing the waste absorption ability of the environment. Waste can be classified into several categories based on its physical state (solid, liquid, gas), use, material type (glass, paper, wood), safety level, and so on. Physical state is the most popular waste classification among these categories. Solid wastes such as paper, glass, food, plastic (ca. 10 %), and metal are more frequently concerned in the community than other substances (Fig. 1). Recently, plastic waste has gained more attention compared to other types of waste, due to its detrimental impacts on the environment and humans (Al-Mosawi et al., 2017; Sokolova et al., 2023).

Discarded plastic debris can accumulate in various natural habitats. Therefore, plastic waste pushes not only the environment but also humans into a critical stage. After being exposed to ultraviolet radiation, plastic debris undergoes a photo-oxidation process, causing it to become brittle (Gunawardhana et al., 2023). The fragmented plastic circulates in different environments and as a result, numerous species are affected by both its chemical and physical impacts. The first encounter of plastics in wildlife, plastic debris in the guts of seabirds, was reported in 1960 (Suaria et al., 2020). The current evidence indicates a growing impact related to public health issues arising from the present use of plastics. Plasticizers, flame retardants, stabilizers, antioxidants, and other additives are added during the manufacturing of plastics to gain specific properties. Consequently, harmful chemicals such as phthalates, bisphenol A (BPA), and polybrominated diphenyl ethers (PBDEs) can be directly transferred to the human body from plastics.

We summarized, analyzed, and synthesized existing research on particular topics such as plastic, plastic waste, environmental impacts, and waste management. This article differentiates itself from other existing works in this field (e.g., Khoaele et al., 2023; Kibria et al., 2023; Tiwari et al., 2023), along with an exploration of the efficacy of current policies and incorporating technological advancements in recycling and waste management. In addition, this article systematically reviewed the characteristics and uniqueness of plastic, classification, waste generation, environmental and public impacts, disposal methods, and rules and regulations for plastic waste management. Therefore, this study investigated the collective global initiatives aimed at fostering a plastic-free world. By scrutinizing these diverse aspects, this review aims to provide a nuanced understanding of the multifaceted challenges posed by plastic waste, and the concerted efforts underway to address them on a global scale.

2. Plastics

2.1. Plastic production, properties, applications, benefits, and consumption

Humans have passed through several eras, and some of the periods have been named after the materials that made remarkable implements and other necessities, like the Stone Age, Bronze Age, and Iron Age. History has repeated itself, as now we are living in the Plastic Age. The word plastic was coined from the Greek word 'Plasticos', which means to be able to form. Plastic is widely known as a synthetic polymer developed by polymerization of monomers extracted from petrochemicals and combined with other chemicals (Torres-Agullo et al., 2021). Monomers are the repeating units of long-chain polymers that are composed of carbon, hydrogen, and oxygen held together by covalent bonds. Ethylene and propylene are common examples of lightweight monomers that are widely used in plastic manufacturing (Hassan et al., 2022). Throughout the polymerization process, certain chemical groups may be omitted from each monomer, resulting in polymers inconsistently preserving the chemical properties or reactivity of the original monomer unit. Polymerization can be 'addition' (combining monomers while maintaining their original structure) or 'condensation' (monomer

structure changes during the process and forms byproducts such as water). Specific chemicals used in plastics vary, depending on the type of plastic manufactured or production process. Plastic production uses several additives like fillers, plasticizers, pigments, foaming agents, processing aids, lubricants, heat stabilizers, acid scavengers, antioxidants, UV stabilizers, flame retardants, and some antistatic agents in different quantities to control different functions like improve properties, ease of processing, increase durability, enhance performance, and appearance.

Presently, the industrial sectors actively explore biomaterials as alternatives to petroleum-based traditional plastic materials (Khoaele et al., 2023; Tiwari et al., 2023). These bioplastics consist either entirely or partially from natural resources such as vegetable oils and starches (plant-based products). Bioplastics have the potential to decrease environmental impacts, specifically reducing carbon footprint, and limiting greenhouse gas emissions associated with polymer utilization. The development of modern bioplastics gained momentum in the 1980s, driven by a desire to minimize landfill waste (Shah et al., 2021). Therefore, the primary objective in the production of bioplastics is to create an environmentally friendly product that matches the quality of conventional plastics. Even though bioplastics account for only 1 % of total plastic production (Narancic et al., 2020), the anticipated expansion of the global bioplastics and biopolymers market is poised to increase from USD 10.5 billion in 2020 to reach USD 27.9 billion by 2025 (Degli et al., 2021).

Plastics have several valuable properties such as being lightweight, toughness, high strength, good stiffness, versatile fabrication, design capabilities, good insulation, a poor conductor of electricity/heat, and corrosion resistance (Ilyas et al., 2018; Pan et al., 2020). Synthetic semi-aromatic plastics (e.g., polyethylene terephthalate: PET and polybutylene terephthalate: PBT) have excellent physio-chemical properties such as high heat distortion temperatures, good mechanical strengths, excellent chemical resistance, and electrical insulation. Properties such as high glass transition temperature (T_g) and high melting temperature (T_m) indicate their ability to maintain shape and strength at elevated temperatures (Wu et al., 2015). The low manufacturing cost, malleability, and processing abilities are also important factors in plastic becoming an important material in every human life.

Accordingly, those properties enhance the usage of plastics, especially during the Second World War with the invention of nylon (Kedzierski et al., 2020). Plastics have many ways to shape life by being involved in every field such as agriculture, construction, transportation, heat insulation, packing, manufacturing, electronics, furniture, toy and leisure items, automobiles, and medicine (Oehlmann et al., 2009; Pan et al., 2020). Currently, 85 % of medical devices such as intravenous bags, disposable syringes, sterile packaging, and joint replacement are manufactured with plastics due to their lightweight and biocompatibility properties (Ilyas et al., 2018). The advantages of plastics can be identified as (i) environmental benefits (e.g., PET beverage packing materials reduce energy consumption and greenhouse gas emission compared to glass and metal packing, and lighter plastic composites in aircraft result in low fuel consumption and easy assembly), (ii) social benefits (e.g., supply and storage of clean drinking water by installing a range of diverse water controlling and distribution systems, and food packaging allow safe and time-dependent storage of fresh products in controlling temperature/pressure conditions), and (iii) economic benefits (e.g., replace automobile parts with low-cost plastic materials).

Traditional materials like glass, wood, metal, and paper gradually being replaced with plastics due to the shortage of raw materials in the modern era (Chandran et al., 2020). Accordingly, global plastic production has grown to match the current plastic consumption pattern (Panda et al., 2010). In addition, plastic production is expected to double in the next 20 years with the present growth rate (Lebreton and Andrady, 2019). China became the world's largest plastic producer in 2010 by surpassing Europe (Huang et al., 2022). China is still considered

the world's largest producer (60 MT out of 359 MT, in 2018) and consumer of plastic. Furthermore, China produces one-third of plastics in global production (Pan et al., 2020). However, western countries are considered the biggest per capita consumers compared to Asian countries (Fig. 1).

2.2. Classification of plastics

Plastics can be classified based on various facts such as origin, structure, molecular forces, polymerization mechanism, temperature-dependent behavior, and preservative techniques. Origin-based classification divides plastics into three subdomains: natural, semi-synthetic, and synthetic. Natural polymers date back to the prehistoric period. Polymers (e.g., lignin, chitin, and starch), which are extracted from natural sources such as horn, amber and tortoiseshell, are known as natural plastics. Synthetic polymers are derived from petrochemicals like silicone, polyethylene, polystyrene, and nylon (Khan et al., 2019; Napper and Thompson, 2020). Semi-synthetic plastics are chemically modified natural polymeric materials like cellulose-based plastics (Khan et al., 2019). The classification based on structure can be divided into linear, branched, or cross-linked. In addition, molecular-based classification can be divided into four forms: elastomers, fiber, thermoplastic, and thermosetting plastics (Fig. 2).

The alternative methods can also applied for specific classification of plastics. For example, the classification based on physio-chemical properties and applications is mainly focused on thermoplastics and thermosetting plastics. Fig. 2 shows the specific classification of plastics based on physio-chemical properties and applications, material origin and biodegradability, life cycle, and international standards. Based on the physio-chemical properties before and after heat treatment, plastics can be divided into thermoplastic and thermosetting plastic. Thermoplastics can melt into any shape while heating and solidify during cooling. Accordingly, thermoplastics can repeatedly deform with typical plasticity. Thermoplastics contribute to the total plastic consumption by 80 % (Al-Salem et al., 2009). Thermosetting plastics decompose directly with continuous heating, instead of deforming (Chen et al., 2019). Based on applications, plastics can be divided into general, engineering, and functional plastics. General plastic has common properties for a wide range of applications. Engineering plastic has excellent comprehensive properties (i.e., stiffness, toughness, and temperature resistance). Therefore, engineering plastics are used as structural engineering materials to replace metal. Functional plastic has special qualities like electrical conductivity, radiation resistance, corrosion resistance, and magnetic permeability (Pan et al., 2020).

According to the material origin and biodegradability characteristics, plastics can be divided into four subgroups (Fig. 2). The main difference between non-biodegradability and biodegradability is their decomposition ability in nature. For example, biodegradable plastics decompose naturally, whereas non-biodegradable plastics remain unchanged for a long period of time (Chandran et al., 2020). Plastics can be further divided based on their life cycle as single-use plastic (the main reason for high pollution) and reusable plastic (McGain et al., 2010). Apart from the above traditional classifications, the Society of the Plastic Industry (SPI) introduced a number-coding system for plastics (Fig. 2). In this classification, numbers from 1 to 7 have been assigned for different plastic types in the following order: 1 for polyethylene terephthalate (PETE), 2 for high-density polyethylene (HDPE), 3 for polyvinyl chloride (PVC), 4 for low-density polyethylene (LDPE), 5 for polypropylene (PP), 6 for polystyrene (PS), and 7 for other types of plastics (Fig. 2). This classification helps consumers and recyclers to simplify the recycling process. Plastics can also be categorized as either bio-based or fossil-based materials. Bioplastics can further be divided into sub-units based on degradation: biodegradable (bio-based), biodegradable (petrochemical-based), and non-biodegradable (bio-based).

2.3. Plastic as a waste material

Plastic waste has become a universal problem, as most plastic waste is difficult to degrade naturally. In particular, most developing countries have no advanced technological facilities and no proper rules and regulations on plastic production, use, and waste management (Eze et al., 2021). Since plastics pollute both terrestrial and marine ecosystems, the United Nations added the statement, 'Plastic is one of the substantial environmental threats faced by the world' (Khan et al., 2019; Borrelle et al., 2020). At present, the transformation of plastics from useful 'objects' to bothersome 'waste' has become a trending topic in the world. The first discussion on plastic waste began in the 1980s (Fig. 1) and continued afterward (Kedzierski et al., 2020). The generation of plastic waste in the world increased gradually due to three main reasons: (i) the ability of plastic to replace traditional materials such as ceramic, wood, and glass, (ii) population growth (e.g., world population increased sharply from 2.5 billion to 7.7 billion from 1950 to 2019), and (iii) easy access to consumer society. Most people migrated to developed areas from the countryside, and the population in urban areas increased from 751 million to 4.2 billion. This population is known as easy access to consumer society (Kedzierski et al., 2020).

The short life-cycle of plastic (i.e., single-use material) increases plastic accumulation in the environment. Roughly two-thirds of global plastics have a short life cycle which is less than a month (Panda et al., 2010). The estimated global plastic waste generation was around 6.30 billion tons from 1950 to 2015, and about 80 % of plastics accumulate in the natural environment. Liang et al. (2021) calculated the plastic waste generation in 2016 to be around 242 million tons. The degradation rates of plastic range from 100 to 1000 years. Consequently, the first invented plastics may still exist in nature (Welden, 2020). Based on the prediction by Stubbins et al. (2021), plastic waste accumulation by 2035 would be equal to the amount of fish in the oceans. Those plastics exist in nature in different particle sizes and are known as macro (>25 mm), meso (5 – 25 mm), large micro (1 – 5 mm), small micro (20 μ m – 1 mm), and nano (1 – 1000 nm) plastics (Gabbott et al., 2020; Hurley et al., 2020; Chen et al., 2021). Sources of plastic waste can also be categorized based on applications (e.g., bottles/caps/lids: 14.9 %, PET bottles: 12.5 %, supermarket bags/retail bags: 9.3 %, food bags: 6.5 %, food containers: 2.1 %, etc.) and origin (e.g., municipal solid waste and agro-industrial waste) (Oehlmann et al., 2009; Geyer et al., 2017; Egun and Egbayiro, 2020).

Plastics exhibit limited natural degradation in the environment due to their high stability and durability. Plastic degradation depends on environmental factors like sunlight, heat, chemical/biological activities, and physical characteristics like polymer size, density, and molecular weight. Weathering and degradation of plastics fall into four categories: physical/mechanical degradation, photodegradation/photo-oxidative degradation, chemical and thermal degradation, and biological degradation. Physical/mechanical processes (wave action and the panning process in coastal environment) involve bulk structure alterations, such as cracking and embrittlement. This process promotes chemical changes that occur at the molecular level, including polymer oxidation and the formation of short-chain molecules. Photodegradation/photo-oxidative degradation (i.e., decomposition of plastics under exposure to ultraviolet (UV) radiation) initiates the breakdown of polymer chains/reducing molecular weight and producing free radicals (Gunawardhana et al., 2023). Chemical degradation (corrosive chemicals including acids such as nitric, sulfuric and hydrochloric, and atmospheric pollutants) and thermal degradation (chemical changes affecting heat/temperature) can break and oxidize the polymer chains of plastic. Microorganisms like bacteria, fungi, and yeasts can change the physical and chemical properties of plastics by metabolizing the polymer carbon into carbon dioxide or incorporating bio-molecules in a natural degradation. This process is known as biological degradation.

Bioplastics also presented new challenges in waste management. For example, some of the existing bioplastics in the market are not biodegradable (Rahman and Bhoi, 2021). However, microorganisms can

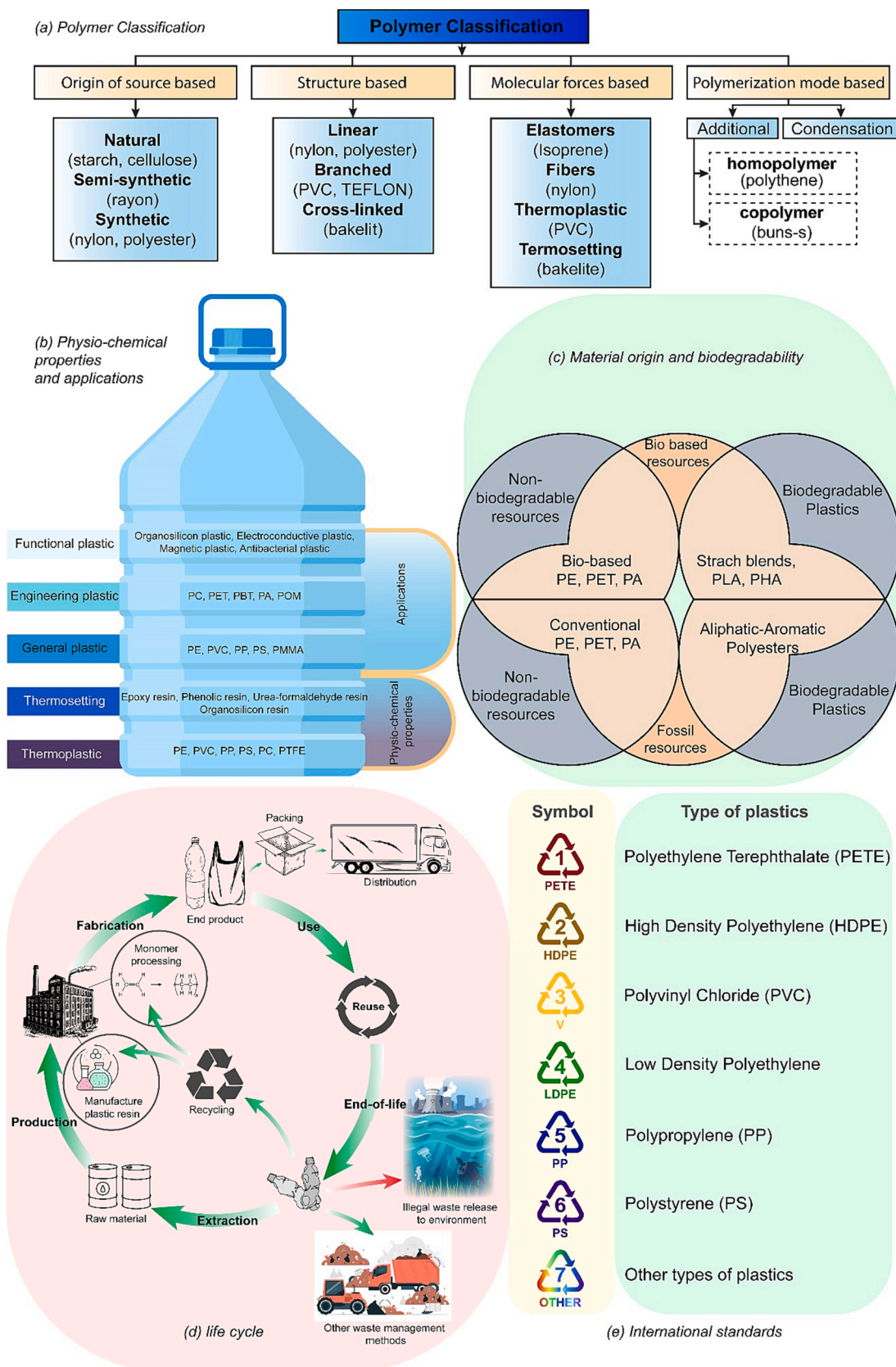


Fig. 2. Classification of plastics.

break down biodegradable bioplastics into carbon dioxide, mineral salt, and water in the presence of aerobic and feasible environmental conditions (Wojnowska-Baryła et al., 2021).

3. Plastic wastes in environments and its impacts

Today, the world produces a remarkable amount of plastic waste as a result of mismanagement, unproductive waste management techniques, and their continuous release (Hu et al., 2019). It is estimated that the world has released around 5800 Mt of mismanaged plastic waste into the environment by the end of 2015 (Geyer et al., 2017). These plastics remain in the environment for a long time/over centuries due to their stability, durability, and their recalcitrant nature to biodegradation, followed by high molecular weight, complex 3D structure, and hydrophobic nature (Wang et al., 2020; Huang et al., 2022). Plastics undergo different weathering processes in the environment, such as photo-degradation, thermal oxidation, hydrolysis, biodegradation, and fragmentation (Wang et al., 2021a,b). However, the ubiquitous presence of plastics spans diverse ecosystems, from deserts to farms, and mountaintops to oceans (Cózar et al., 2014; Gabbott et al., 2020).

3.1. Plastics in terrestrial and marine environments

Since plastics are mainly manufactured, used, and disposed of on land, a high amount of waste accumulates in the terrestrial environment (Hurley et al., 2020). For example, agricultural plastics such as poly-tunnel plastic films, wrappings, and packing materials are highly available as plastic wastes. Those plastic particles break into

microplastics (Fig. 3) and have the ability to remain in the soil for up to 15 years. In addition, illegal dumping activity and direct input of waste increase plastic accumulation on land (Hurley et al., 2020). The terrestrial environment can act as a plastic sink and bare them up to a certain level (Fig. 3). Human/anthropogenic, wind (aeolian), and water (fluvial) activities transport plastic litter. Apart from land, freshwater systems such as rivers, dams, lakes, and urban drainage networks have been surpassed by plastic debris (Azevedo-Santos et al., 2021).

Freshwater ecosystems serve as the primary receptacle for various pollutants released within a watershed, owing to the natural positioning in valleys and lower elevation terrains including floodplains (Azevedo-Santos et al., 2021). Rivers are identified as a primary source to carry plastics in the marine environment (Hurley et al., 2020; Chen et al., 2021). Based on the United Nations Environmental Programme (UNEP), nearly 1,000 rivers are responsible for almost 80 % of the annual riverine plastic emissions worldwide, accounting for a range between 0.8 and 2.7 million tons per year. Wind is the main transportation agent for small-sized plastic particles (i.e., micro- and nanoplastics) in urban areas. Atmospheric circulations can spread micro- and nanoplastics across a wide space (Azevedo-Santos et al., 2021). Accordingly, particle size is important in the mobilization and accumulation of plastics in the atmosphere.

Eriksen et al. (2014) identified that the amounts of plastic entering the ocean depend on three principal variables, including watershed outfalls, population density, and maritime activity (Fig. 3). Every year, about 8 to 10 million tons of plastics, including both macro-plastics and microplastics, leak into the ocean and account for 80 % of all marine pollution. Plastic wastes kill more than 100,000 marine species each

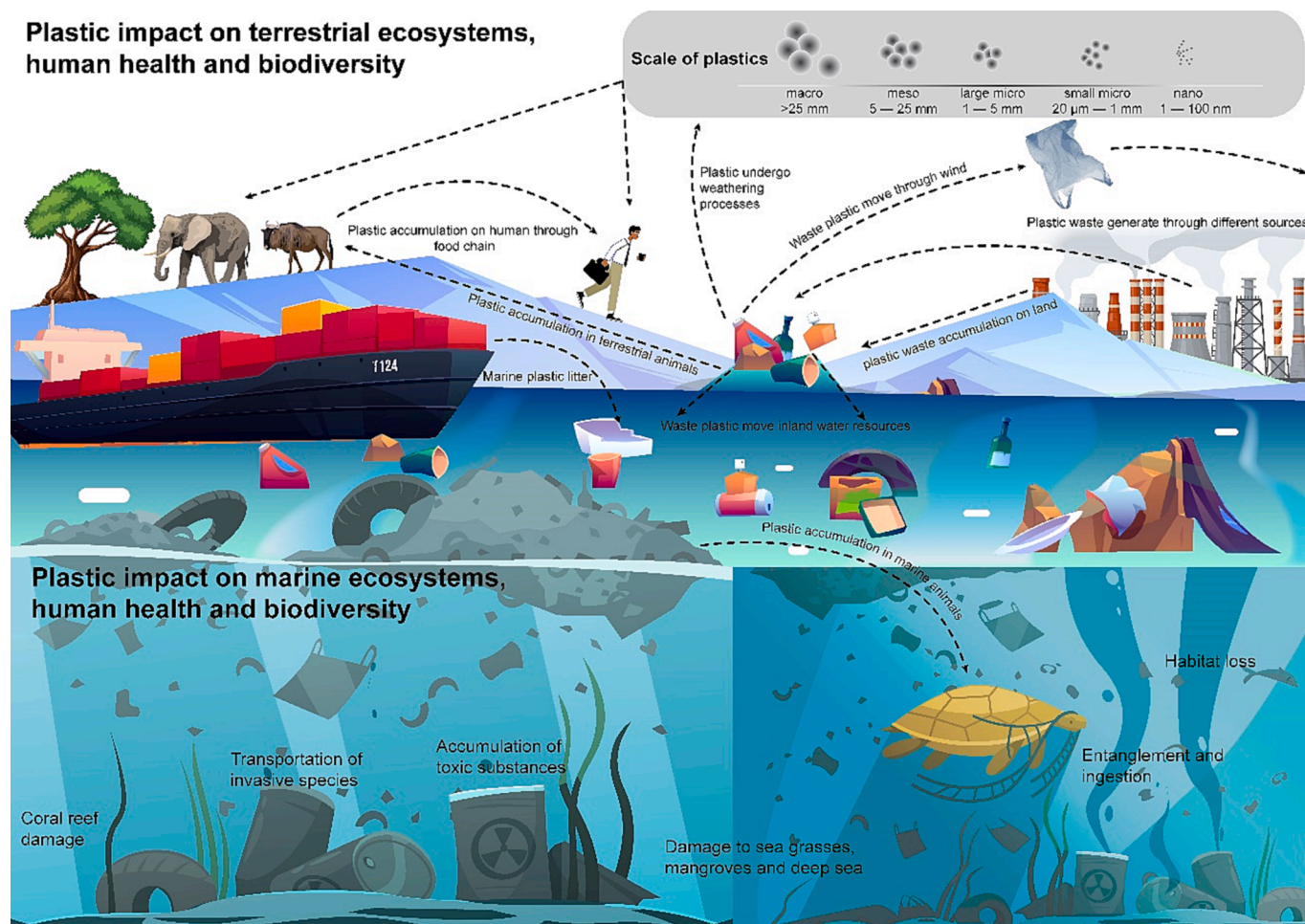


Fig. 3. Plastics in terrestrial and marine environments.

year, covering all areas from coastlines to the deep ocean (Cózar et al., 2014; van Sebillie et al., 2015). Maritime accidents like the MV X-Press Pearl accident release tons of plastics into the Indian Ocean (Perera et al., 2022; Ratnayake and Perera, 2022). Buoyancy, durability, wind, and oceanic currents allow the spreading of plastic waste over a large area (Eriksen et al., 2014; Jambeck et al., 2015). For example, trans-boundary impacts were identified in the MV X-Press Pearl accident due to the spreading of plastic pellets across Indian Ocean coastlines from Indonesia and Malaysia to Somalia (Perera et al., 2022; Ratnayake and Perera, 2022). In the marine environment, plastic wastes are categorized into three groups: (i) floating debris/garbage patches (aggregate with gyres, e.g., Great Pacific garbage patch), (ii) seafloor plastics (a common phenomenon in almost all major oceanic systems and depending on seasonality, local and global currents, bathymetry, and local sources), and (iii) shoreline plastics (plastic wastes on the coast, depending on circulation and seasonal weathering patterns, coastal tourism, and socio-economic backgrounds such as urbanized areas and changes in social habits) (Welden, 2020).

3.2. Plastics in the atmosphere

Airborne microplastics consist of both natural and synthetic polymers. The predominant size of airborne microplastics ranges from 20 to 500 μm , with a majority around 20 μm (Wang et al., 2021a,b). Airborne microplastics contain different colors (black, and white), and the shapes are different such as fiber (the most prevalent), film, fragment, foam, granule, and sphere (Wang et al., 2021a,b). Synthetic textiles, especially small fibers used in clothing, serve as a significant source of airborne microplastics (Huang et al., 2020). For example, 1 g acrylic fabric produces more than 1100 airborne microplastics (Wang et al., 2021a,b). These fibers can be released into the atmosphere during activities like wearing clothes or drying, especially from soft furnishings like carpets and curtains (Fig. 4). Airborne microplastics can also have alternative sources, including the degradation of larger plastic items and industrial

emissions. In general, the amount of airborne microplastic indoors is much higher than the outdoor concentration (Wang et al., 2021a,b). The dispersion and subsequent deposition of airborne particles in atmospheric fallout are influenced by factors like rainfall, wind, pollution concentration, humidity, local conditions, and particle size (Fig. 4). The wind serves as the force lifting microplastics from the ground into the air, facilitating their transport to remote areas. Upon deposition, these airborne plastics become a source of contamination in both terrestrial and aquatic ecosystems. Consequently, this dynamic cycle exchanges plastics among the air, terrestrial, and aquatic environments, contributing to their presence in the overall environmental landscape.

3.3. Impact of plastic debris on biodiversity

Plastic debris causes harm to biodiversity through physical, chemical, and biological impacts (Wang and Qian, 2021).

3.3.1. Physical and chemical impacts

Animals like reptiles (e.g., mostly snakes) and birds can be trapped by plastic nets (commonly applied in farm fields, gardens, and aquatic farms) (Stuart et al., 2001; Townsend, 2011; Avery-Gomm et al., 2013). Animals may encounter plastics (mainly microplastics) in various environmental settings, resulting in ingestion (Fig. 3). Accordingly, the accumulation of plastics in digestion systems causes several health issues for ruminants camel, goat, and sheep, such as internal injuries, blocking of the intestinal tract or gastric enzyme secretion, diminished feeding stimulus, failure to absorb volatile fatty acids, reduction of animal fattening, and lowering of steroid hormone levels (Omidi et al., 2012; Hurley et al., 2020). Therefore, plastic ingestion significantly influences feeding behavior. In marine environments, plastic debris reduces the lighting and oxygen levels of underlying water, and this reduces the biodiversity of marine habitats (Pawar et al., 2016). Ocean currents transport plastic from fishing gear, nets, and other plastic substances to shallow habitats and then damage and degrade coral reefs

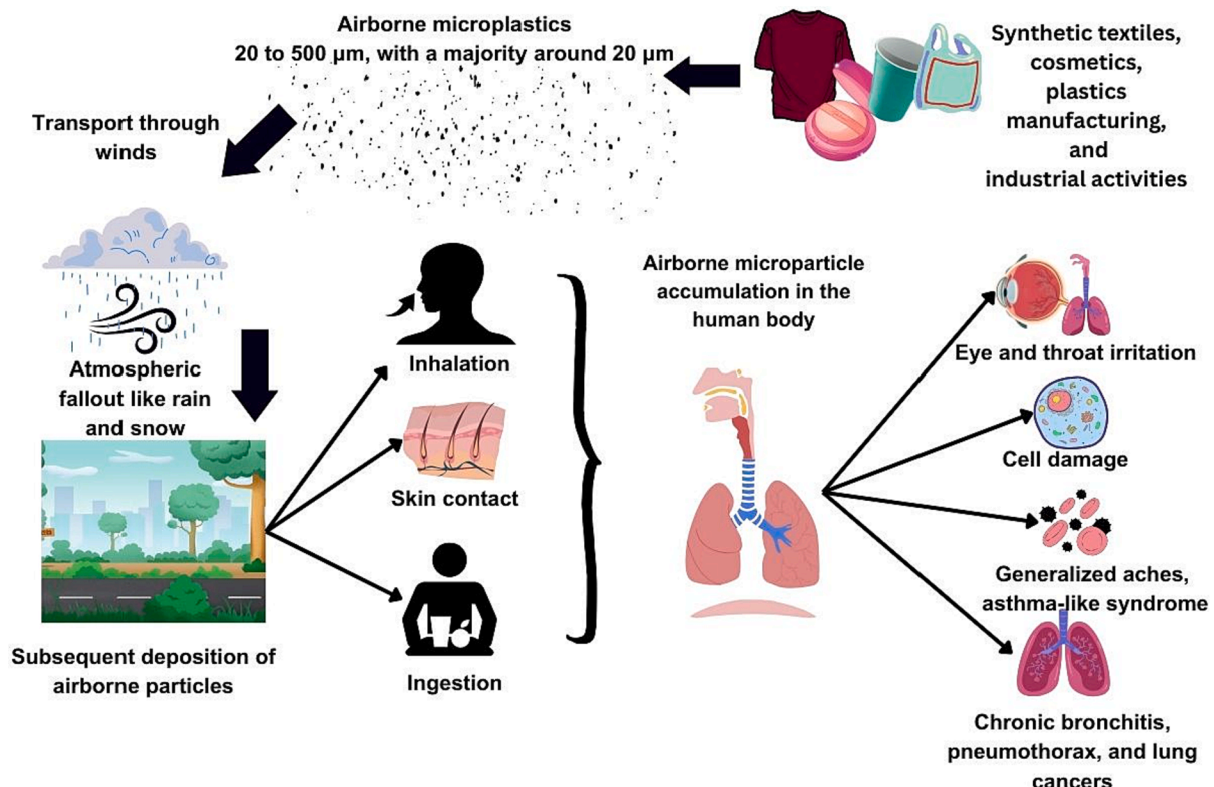


Fig. 4. Plastics in the atmosphere and impacts on public health.

(Chiappone et al., 2005). The blanketing effect of plastic sheeting causes anoxia and hypoxia.

Chemicals associated with plastic such as, bisphenol A, phthalates, and heavy metals, can accumulate on organisms through food webs, and it causes negative impacts like oxidative stress, cancer, and endocrine disruption (Wang and Qian, 2021). The leaching rates of plastic-related chemicals are high in marine environments due to ultraviolet radiation (UV) breakdown, weathering, and mechanical forces (Gunawardhana et al., 2023). These chemicals first leach into the environment/seawater and then reach biota by ingestion. For example, polychlorinated biphenyls (PCBs) enter the food chains and lead to reproduction disorders or even death (Avery-Gomm et al., 2013; Pawar et al., 2016). These physical and chemical impacts lead to biological impacts.

3.3.2. Biological impacts

Microplastics can cause biological impacts on living organisms by controlling their growth, reproduction ability, and long-term toxicity (Zhang et al., 2023). Microplastics derive from polyvinyl chloride (PVC) disturb growth, oxidative damage, oviposition, and biological enzyme activity (Espinosa et al., 2017). Browne et al. (2013) identified biological impacts on lugworms due to the incorporation of microplastics and additives into their gut tissues. In addition, insects like *Culex* mosquitoes show adverse interactions with microplastics (Hurley et al., 2020). Microplastics also cause energy and lipid metabolism disorders and oxidative stress in mice.

According to the literature data, plastic debris globally affects many marine species, including about 43 % of marine mammals, 44 % of seabirds, and 86 % of sea turtles and various fish and crustacean species (Thushari and Senevirathna, 2020). The major biological impacts of plastic waste in marine and other aquatic environments can be summarized as ingesting, entangling, impaired movement, habitat loss and feeding, low reproduction, ulcer, laceration, and finally, death (Laist, 1997; Schnurr et al., 2018). For example, ingesting causes internal blockages and injuries, low stomach capacity, and hinders growth. Entangling results in strangulation, reduction of feeding efficiency, and some cases drowning. Oceanic turtles consume floating plastic debris including plastic bags due to a misunderstanding of those as their gelatinous prey/jellyfish which is the primary food source of sea turtles (Pawar et al., 2016). Seabirds eat various types of plastic debris including balloons, plastic pellets, and hard bits of plastic by thinking of food. Accordingly, low body weight, inhibited fat deposition, reduced reproductive ability, low fitness, and digestive system damage are the main problems associated with seabirds (Pawar et al., 2016; Alabi et al., 2019).

Microplastics resemble phytoplankton and sometimes plastics mix with the desired foods of fish and cetaceans (Boerger et al., 2010). Cetaceans face direct mortality or may become weak and prone to predation or disease due to the effect of plastic debris (Sigler, 2014). A study in North Pacific Central Gyre estimated that 35 % of fishes out of 670 contained a total of 1375 plastic pieces in their bodies and approximately 2.5 pieces per fish. Most of those plastics are in plankton colors (the primary food source of fish) such as blue, white, and clear (Davison and Asch, 2011; Sigler, 2014). Therefore, plastic fragments can accumulate in predators within the food chain (Teuten et al., 2009). In particular, nanoplastics derived from polystyrene (PS) decrease the filter-feeding rate of blue mussels and impact the cellular and organizational levels (Rossi et al., 2014). Furthermore, microplastics of polystyrene reduce the number and size of egg cells, sperm motility, and larval number for oysters (Sussarellu et al., 2016), and also reduce the growth rate of *Gammarus pulex* (Redondo-Hasselerharm et al., 2018). Moreover, micro- and nanoplastics affect the temperature-related biological processes, and cause the change in the sex ratio of turtle eggs during incubation (Hu et al., 2019).

Apart from fauna, plastics interact with terrestrial and aquatic flora. Preliminary laboratory observations suggest that plastic traces available in the soil can negatively affect aquatic plants by reducing shoot and

root length (Hurley et al., 2020).

3.3.2.1. Entanglement and ingestion. Entanglement occurs when animals come across loops or openings of debris and are entrapped, causing suffocation, drowning, strangulation, an inability to escape from predators, diminished mobility, migration activities, or death. The offspring of various marine species could face consequences from fishing nets, and other plastic artifacts, such as bands, collars, and straps, discarded by maritime activities. These items may become entangled around their necks or bodies, constricting them, and posing a strangulation risk as the animals grow (Thushari and Senevirathna, 2020). Ingestion is closely linked to individual organisms. Notably, fulmar, seals, sea lions, puffin, albatross, right whales, green sea turtles, hawksbill turtles, and greater shearwater are negatively impacted due to entanglement and ingestion. The impact of ingesting marine debris on wildlife health manifests through two interconnected categories: physical and physiological effects (Jacob et al., 2020). The physical effects encompass (i) lacerations and lesions: occurring when sharp debris punctures the lining of the digestive system, leading to ulceration, lesions, infection, and inflammation, (ii) blockage: wherein indigestible sheets and plastic bags become lodged in the gastric system, exposing organs to digestive fluids, and causing a false sense of satiety, and (iii) retention: signifying the prolonged presence of debris in the digestive tract. All these physical effects eventually culminate in physiological consequences, including nutritional, developmental, immunological, and toxicological impacts. In addition, the chemical toxicants associated with plastic affect animal development, reproductive cycles, and population dynamics, potentially yielding long-term consequences.

3.3.2.2. Transportation of invasive species. Invasive species have a higher ability to settle outside their usual habitats, proliferate uncontrollably, and pose a threat to the well-being of native species, and the health of the ecosystem. Another vector of non-native species is drafting through debris. The resilience, and buoyancy of plastics create long-range transport of species/materials to ecosystems such as polar environments where they are not native (Andrade et al., 2021). Previous studies indicate that microorganisms from the *Vibrio* family are capable of rafting on plastics, and microplastics (Napper and Thompson, 2020). Additionally, plastics may provide habitats, as evidenced by the insect *Halobates micans*, which has been observed using plastic litter as ovipositional sites (Thushari and Senevirathna, 2020). Such invasions by invasive species are highly dangerous to endangered ecosystems.

3.3.2.3. Accumulation of toxic substances. Plastic pollution is also associated with chemical hazards, especially those containing endocrine-disrupting chemicals. In addition, hydrophobic chemicals adsorb environmental pollution (Gallo et al., 2018). These chemicals can be released directly into the digestive system of organisms or leach into the environment. The potential impact includes impaired reproduction, low birth rates, biodiversity loss, thyroid issues, metabolism disruption, and an increased risk of hormone-sensitive cancers. The critical-sensitive periods for these effects include embryonic and developmental stages, with adverse outcomes observed at extremely low concentrations (Gallo et al., 2018).

3.3.2.4. Impacts on coral reef. Different sizes of plastic particles interact with coral reefs in different ways and cause entanglement, feeding, and covering/asphyxiation. Corals often consume microplastics mistaken as zooplankton due to their size and other similarities (John et al., 2021). Ingested microplastics cause severe tissue damage, break down their immune system, reduce food intake, and lead to slow growth, necrosis, and bleaching. The toxicity associated with plastics can also lead to photosynthesis issues (*Acropora* species, *Seriatopora hystrix*), calcification effects (*Pseudodiploria clivosa*, *Acropora cervicornis*), decrease enzyme activities (*Tubastrea aurea*), and microplastics cluster formation

in gastrovascular tissues (*Porites porites*) (John et al., 2021). Previous experiments revealed the possible long-term impacts of the feeding behavior of the corals. For example, according to Mendrik et al. (2021), the photosynthetic activities of *Acropora* species decreased by approximately 41 % within 12 days.

3.3.2.5. Impact on seagrasses, mangroves, and deep seafloor ecosystem. Plastic pollution influences the nitrogen release from seagrass litters, thus potentially influencing coastal nutrient and biogeochemical cycles. According to Walther and Bergmann (2022), the presence of plastic pollution decreased the decomposition rate of eelgrass by 36 %. In addition, seagrasses are one of the sinkers of microplastics in coastal areas (Walther and Bergmann, 2022). Mangrove forests can significantly trap plastic debris due to their intricate aerial root systems. For example, in 2010, mangrove forests along the Red Sea and Arabian Gulf exhibited a litter density of 66 items per 100 m², with over 90 % comprising plastics (Martin et al., 2019). The largest single mangrove forest in the world, Sundarbans in Bangladesh and India, is under immense threat of plastic pollution. The Sundarbans has turned into a dumping ground for plastic waste that discharges through rivers (Adyel and Macreadie, 2021).

The deep seafloor also acts as a potential sink for plastic litter such as municipal waste, approximately 50 % of plastics from municipal waste are denser than seawater, directly sinking to the seafloor. In general, plastics gradually descend to the seafloor over time, influenced by ocean currents, material degradation, and ballasting processes. Furthermore, microplastic contamination has been recorded on the deep seafloor (Kane and Clare, 2019). Microplastics exhibit faster sinking in the deep oceans due to their heightened surface-to-volume ratio, leading to increased fouling (Kane et al., 2020). This effect is particularly pronounced when these particles are incorporated into aggregates and feces.

3.4. Impact of plastic debris on public health

Beyond the environmental boundaries, plastics interrupt human lives as well (Fig. 5). The available chemical additives in plastics such as Bisphenol A (BPA), phthalates, and Brominated Flame Retardants (BFRs) can cause critical damage to public health (Pawar et al., 2016). Bisphenol A, a carbon-based synthetic plasticizer, consists of two unsaturated phenolic rings attached to bridging carbon. It is known as an endocrine disruptor and BPA is used in large quantities as a plasticizer in various commercial products (e.g., food and drink packages, compact discs, plastic cans, bottle tops, sunglasses, medical devices, and water pipes), due to its high durability, heat-resistant ability and shatter resistance properties (Chouhan et al., 2014; Rezg et al., 2014). People are exposed to BPA through various sources such as air, dust, and water, but the main source is diet (Kang et al., 2006; Katyal et al., 2020). The breakdown of BPA occurs due to hydrolysis under high temperatures and various pH conditions (Chouhan et al., 2014; Campanale et al., 2020). The presence of BPA is detectable in the blood of pregnant women, amniotic fluid, placental tissue, and cord blood, indicating fetal exposure. Young infants are often exposed to BPA through the use of feeding bottles containing plastics (Kumar, 2018). Bisphenol A can cause serious issues in human health like cardiovascular disease, chronic kidney disease, birth defect and developmental disorder, cancer, and chronic respiratory disease (Rezgi et al., 2014; Campanale et al., 2020).

Phthalates are used as plasticizers in various plastic products including personal care products, toys, and food packing. Phthalates enter the human body through ingestion, inhalation, and dermal contact (Wang and Qian, 2021). There are various types of phthalate compounds such as diethyl phthalate, di-*n*-butyl phthalate, butyl benzyl phthalate, di (2-ethylhexyl) phthalate, and those phthalates have different potential health effects (Fig. 5). For example, diethyl phthalate is used in personal care products (e.g., fragrances), and it has potential health effects of reduced growth rate and food consumption. Di-*n*-butyl phthalate is mainly used in nail polish and cosmetics, and it can cause

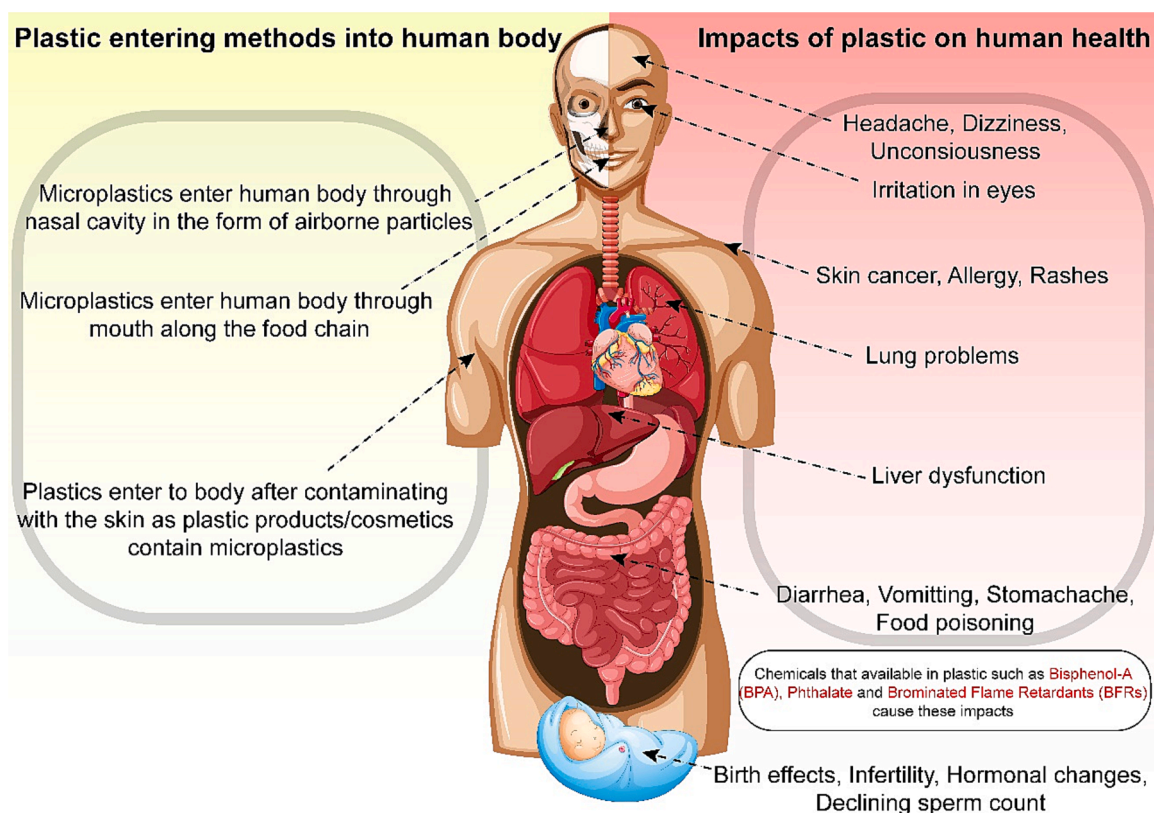


Fig. 5. Plastics entering methods and impacts on human health.

hepatic and renal effects (Hauser and Calafat, 2005).

Brominated Flame Retardants (BFRs) are flame-resistant (i.e., prevent fire damage) chemicals that are used in electronic types of equipment such as televisions, pulse oximeters, monitors, and ventilators (Fig. 5). There are various types of BFRs (e.g., polybrominated diphenyl ethers: PBDEs, tetrabromobisphenol-A: TBBPA, and hexabromocyclododecane: HBCD), and those BFRs can cause serious health issues (Kumar, 2018). For example, polybrominated diphenyl ethers can cause neurological disorders and cancer (Emoyan et al., 2021).

Plastics mainly in the forms of micro- and nanoplastics can easily enter the human body in numerous ways such as airborne and through the food chain, due to their compatibility size (Figs. 4 and 5). Airborne microplastics can enter the human respiratory system through breath (Enyoh et al., 2019; Huang et al., 2020). However, the clearance mechanism, which acts as a defensive system, safeguards the entered airways through the nose during normal breathing, by capturing foreign particles using various methods including sneezing, mucociliary escalator, phagocytosis, and lymphatic transport (Vianello et al., 2019). Several studies discovered fibers in the deep lung (e.g., Enyoh et al., 2019), and these microplastics can migrate to lymph nodes, other tissues, and organs through the bloodstream due to their hydrophobic nature (Fig. 4). Therefore, it causes severe health issues such as eye and throat irritation, shortness of breath, cough, chest tightness, phlegm, wheezing, frequent fevers, generalized aches, asthma-like syndrome, chronic bronchitis, pneumothorax, and lung cancers (Enyoh et al., 2019; Wang et al., 2021a,b). Workers in industries such as synthetic textiles and plastics manufacturing have a high potential to contaminate airborne microplastics due to their long-term exposure (Fig. 4). A recent study shows that airborne microplastics may transport SARS-CoV-2 virus (COVID-19) for long distances from the source (Liu and Schauer, 2021).

3.5. Direct socio-economic impacts of plastic waste

Plastic waste in ecosystems has led to several direct social and economic challenges in fisheries, tourism, shipping, human health, and agricultural soils. These challenges have detrimental effects on the economy as an extra necessitating budget for waste removal. For example, macro- and microscopic plastic remnants can change physicochemical and hydrological parameters in agricultural soils, and thus ultimately reduce the soil quality (Wan et al., 2019; Qi et al., 2020). In addition, plastic pollution is directly connected with the fishery and tourism industries. Marine debris significantly blocks fishing stocks, with 86 % of Scottish fishing vessels experiencing restricted catches due to litter in their nets. This issue costs an average of Euro 11.7–13.0 million (USD 12.9–14.3 million) annually, reducing approximately 5 % of the total revenue (Arabi and Nahman, 2020). The depreciation of fisheries resources is also linked to the economic impacts. Plastic pollution has also negatively impacted seafood safety. This decreases the demand and results in economic losses globally. For example, the MV X-Press Pearl accident in 2002 released over 1750 tons of plastics into the Indian Ocean, and contaminated nurdles were found in the gills of seafood species (Perera et al., 2022). Therefore, seafood consumption was notably reduced, which led to economic loss in Sri Lanka (Ratnayake, 2023). Similarly, microplastic contamination is high in aquaculture (Wu et al., 2023). In addition to biological/toxicological impacts, microplastic contamination diminishes the quality and quantity of aquaculture products (Espinosa et al., 2019; Wan et al., 2019; Jacob et al., 2020; Wu et al., 2023). The demand and economic benefits are thus generally reduced for various aquaculture products, encompassing fish, mussels, shrimp, and crabs.

The tourism industry is markedly influenced by the environmental quality of destinations such as beaches. Accordingly, plastic pollution in beaches and marine environments has a detrimental impact on aesthetic value, natural beauty, and ecosystem health. Consequently, the diminished aesthetic and recreational appeal (e.g., snorkeling, turtle/whale

watching), especially in coastal and marine environments, result in a significant reduction in the total number of tourists. The death of corals caused by plastic debris implies the loss of such tourism activities. Similarly, a high concentration of debris on the seabed and coral reefs can profoundly affect diving, as divers avoid heavily polluted locations to prevent entanglement in marine plastics, such as fishing lines (Jacob et al., 2020). Therefore, a decreased number of tourists leads to the loss of job opportunities and is directly intertwined with the negative social aspect (Ratnayake, 2023).

Plastic wastes add direct and indirect costs to the global economy. The direct costs associated with plastic pollution, marine ecosystem degradation (losses in marine tourism, revenue reductions in fisheries/aquaculture), and government cleanup expenditures, whereas the indirect costs encompass public health issues. The regional marine cleanup costs vary in different geographical regions. The comparative analyses indicate that Asia has the highest average costs due to the high emission of waste. In addition, Europe and North America bear the highest revenue loss due to plastic waste (Williams and Rangel-Buitrago, 2019).

3.5.1. Economic analysis

Plastic pollution is responsible for a 1–5 % reduction in human benefits derived from oceans, equating to annual losses of USD 2,500 billion. The costs associated with waste management technologies vary globally: USD 0.003–0.23/kg for mechanical recycling, USD 0.083 for chemical recycling (pyrolysis), USD 0.102 for chemical recycling (gasification), and USD 0.04–0.15 for incineration (Nikiema and Asiedu, 2022). On a national level, waste management costs differ significantly. Shanker et al. (2022) reported that plastic waste management expenditures in India range differently, like dump site disposal (USD 6.02–18.05/ton), landfilling (USD 19/ton), incineration (USD 36–38/ton), mechanical recycling (USD 0.24–1.80/kg), chemical (diesel/fuel) recycling (USD 0.24–0.48/l), and biological recycling (USD 1.08–1.44 per/kg). The Department of Energy reports a USD 7.2 billion expenditure for landfilling in the USA. According to the North America Plastic Recycling Market Report, the USA has spent USD 2.6 billion on recycling and USD 17–24 million on incineration (Law et al., 2020).

4. Plastic and plastic waste management schemes

The world needs a realistic solution to overcome the dark side of plastic and plastic waste. Theoretically, plastic waste management can be divided into three main stages: (i) the production stage (a closed-loop process), (ii) the consumption stage (reduce the consumption), and (iii) the end-of-life stage (reuse, and recycling) (Wichai-utcha and Chavalparit, 2019). The main challenges for plastic recycling are a long time for plastic degradation naturally and high treatment costs or serious environmental problems during unreasonable/simple recycling treatment methods (Pan et al., 2020). Therefore, suitable disposal methods depend on material type, volume, collection, and processing infrastructure (Chandran et al., 2020).

4.1. Biodegradable plastic waste end-of-life options

The degradable rate of biodegradable plastics depends on the biotic and abiotic factors including oxygen availability, humidity temperature, and the presence of microorganisms (Fig. 6).

4.1.1. Biological treatments: Composting and anaerobic digestion

So far, biological methods are the best option for managing biodegradable plastic waste. This process involves several microorganisms such as fungi, bacteria, and Actinomycetes in various environmental conditions (Fig. 6). According to ISO 15270: plastic guidelines for the recovery and recycling of plastic waste, composting is the main biological recycling option for biodegradable plastic. Composting leads to the transfer of organic matter into minerals, biomass, and humus under aerobic conditions, and also releases carbon dioxide, methane, water,

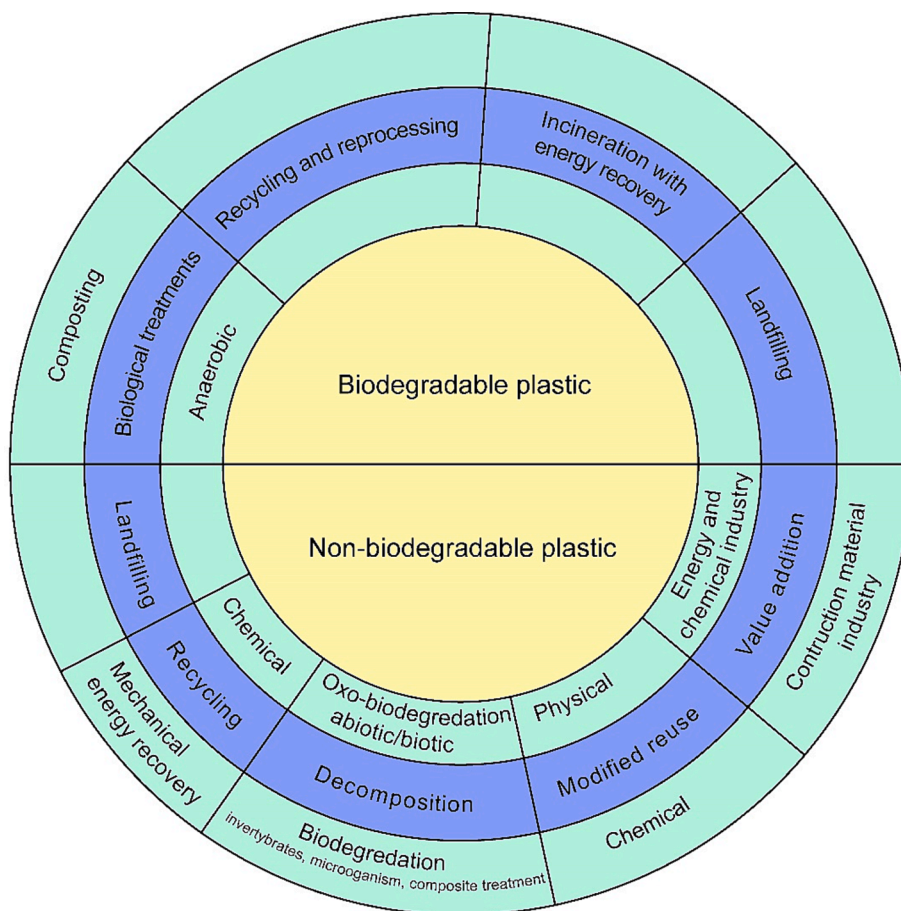


Fig. 6. End-of-life options for biodegradable and non-biodegradable plastic waste.

and heat (Flury and Narayan, 2021). In the industrial composite process, more than 90 % of biodegradable plastics convert into polymer carbon and then into CO_2 within 180 days at a temperature of $58 \pm 2^\circ\text{C}$. The rest of the carbon is incorporated into microbial biomass and CO_2 which is transferred into carbonic acid. Therefore, industrial composting goes through several controlled conditions, whereas domestic composting has variable conditions. However, biodegradation is slow and less effective in domestic composting due to low processing temperatures. This process does not provide suitable nutrients to soil or plants (Haider et al., 2019), and chemical additives of plastics remain in the compost (Chandran et al., 2020; Flury and Narayan, 2021).

In anaerobic digestion, microorganisms convert organic matter into biogas without oxygen, but rich in methane. The anaerobic performance emits less amount of greenhouse gases than composting and incineration (Chandran et al., 2020). Anaerobic digestion process is divided into four stages, such as hydrolysis, acidogenesis, acetogenesis, and methanogenesis (Flury and Narayan, 2021). Anaerobic digestion transforms organic matter into three primary substances: (i) biogas: rich in methane and used as a fuel, (ii) biosolids: microorganisms grown on the organic matter, (iii) liquor: dissolved organic matter used as fertilizers (Saalah et al., 2020).

4.1.2. Recycling and reprocessing

According to the plastic strategy announced by the European Union, it is targeted to reuse and recycle 100 % of plastics by 2030 (Haider et al., 2019). Mechanical recycling of biodegradable plastics undergoes several stages including separating, grinding, washing, drying, regranulating, and compounding, and then recyclates convert into new products (Chandran et al., 2020). Sorting and separating play an important role in biodegradable plastic recycling to prevent

contamination from non-biodegradable plastics, and it also helps to obtain quality recyclates (Chandran et al., 2020). Traditional sorting and separation (e.g., optical and mechanical) methods have limitations to quality separations of biodegradable and commodity plastics due to similar appearance and density. Therefore, new technologies like infrared spectroscopy have been introduced in the sorting and separation processes of mechanical recycling (Haider et al., 2019; Chandran et al., 2020).

Chemical recycling involves the recovery of building block materials, and those recovered materials are used in a new polymerization to produce new materials. Chemical recycling follows two processes: depolymerization (hydrolysis and glycolysis), and thermolyses (gasification and pyrolysis) (Chandran et al., 2020). Some biodegradable plastics like polylactic acid (PLA) can undergo both mechanical and chemical recycling similar to commodity plastics.

4.1.3. Incineration with energy recovery

Since biodegradable plastics contain only carbon, oxygen, and hydrogen atoms, incineration is still used in some energy recovery plants. Incineration is also preferred over land-filling. However, incineration is not the best option for biodegradable plastic disposal. In addition, incineration leads to the release of CO_2 and causes greenhouse effects (Chandran et al., 2020).

4.1.4. Landfilling (Biodegradable plastic waste)

Landfilling is the least favored option for biodegradable plastic waste disposal. Biodegradable plastics from garden and kitchen waste produce methane in landfills due to anaerobic conditions (Fig. 6). Such captured landfill gases can be used as an energy source (Song et al., 2009). However, methane is 25 times stronger greenhouse gas than CO_2 .

Therefore, landfills act as a main methane outlet by contributing to 16 % of the total world's methane production (Chandran et al., 2020).

4.2. Non-biodegradable plastic waste end-of-life options

Non-biodegradable plastics consist of conventional raw materials (e. g., hydrocarbons obtained from crude oil or natural gas) and these plastics take hundreds of years to degrade naturally (Bhagwat et al., 2021; Singh et al., 2022). Accordingly, non-biodegradable plastics are considered one of the highest levels of environmental pollution in the world (Zhang et al., 2021).

4.2.1. Landfilling (Non-biodegradable plastic waste)

The lightweight plastic wastes can easily move in an open environment through air and water sources. Landfilling is thus considered the primary plastic waste disposal method to avoid broader contamination (Huang et al., 2022). In this method, a large quantity of waste can be disposed of through land-filling in a large space often with mound pits or natural pits (Pan et al., 2020). Although a large amount of plastic waste can be disposed of at once, there are several drawbacks including slow decomposition of materials (Rebeiz and Craft, 1995), soil contamination by primary and secondary pollutants such as additives, plasticizers, and pigments (Huang et al., 2022), and plastic wastes becoming permanent garbage (Pan et al., 2020).

4.2.2. Recycling

Recycling involves transforming plastic waste into secondary materials that can be reintegrated into the system, either for reuse in the same capacity or for creating new components and products with comparable or enhanced functionality. This approach aims to effectively 'close the loop' by minimizing waste and maximizing the utilization of recycled materials within the circular economic framework. Recycling has three main types: (i) physical/mechanical, (ii) chemical, and (iii) thermal/energy recycling (Fig. 6). In all types of plastic recycling, sorting/separating is the fundamental step in the processing loop. Several sorting technologies have been introduced in the industry to achieve effective recycling (Al-Salem et al., 2009). These technologies can be mainly divided into manual (visual identification based on shape, color, and plastic trademark), automated dry, and automated wet sorting. Automated dry sorting can be further divided into infrared sorting (use near-infrared wavelength from 600 to 2500 nm), X-ray sorting (identify spectral fingerprint when plastics pass on a conveyor and sort it based on chemical composition), air sorting (0.25 to 0.50 in. flake plastics feed vertically and separate based on weight), electrostatic sorting (fall freely through an electric field and separate based on triboelectric charge), mechanical sorting (separate based on specific gravity, and melting (sorting based on melting point differences in plastics)) (Lange, 2021; Kibria et al., 2023). In addition, automated wet sorting can be further divided into sink float (separate based on density), hydrocyclone (separate suspended materials based on differences in gravity with aqueous solutions), selective dissolution (dissolution of mixed plastics using solvents like *n*-methyl-2-pyrrolidinone, and then precipitate, filtrate and finally evaporate), and hydroglycolysis (polymer molecules convert back to raw monomers and reuse in the manufacture of a new polymer) (Masoumi et al., 2012; Ruj et al., 2015).

4.2.2.1. Mechanical recycling. Mechanical recycling is a physical process of transforming plastic waste into secondary raw materials or products through a series of mechanical steps such as cutting/shredding, contaminant separation, floating, milling, washing and drying, agglutination, extrusion, and quenching (Aznaar et al., 2006). Mechanical recycling is a low-cost method without undergoing substantial changes in the chemical structure of the materials. The process begins with the sorting of plastic waste, which is then cleaned to remove any impurities. The cleaned plastic waste is shredded into small pieces and then melted

down to create pellets or flakes. These pellets or flakes can be used to produce new plastic products. However, mechanical recycling has some limitations. It is not always possible to recycle all types of plastics especially those are contaminated or made of multiple materials. Additionally, the quality of recycled plastic may not be as high as that of virgin plastic. Therefore, mechanically recycled plastics may limit their application.

4.2.2.2. Chemical recycling. Chemical recycling is an advanced process to convert end-of-life plastic materials into small molecules/degradation of polymer bonds (Damayanti et al., 2022). These products are used as feedstock in plastic and petrochemical production (Dogu et al., 2021). Chemical recycling also produces various fuel fractions from plastic waste. Thermolysis is the main chemical recycling method where plastic waste is depolymerized again into monomeric level or to new chemicals under controlled temperature. This process consists of three methods such as: pyrolysis, gasification, and hydrogenation (Al-Salem et al., 2009).

Pyrolysis (i.e., thermal cracking of polymers in inert atmospheres) is responsible for producing clean, high-calorific value gas from plastic waste using different technologies (Table 1). Accordingly, this method converts the hydrocarbon content of waste into gas, which is useful in gas engines, with associated electricity generation, or in boiler applications without the need for flue gas treatment. Fuel, monomers, and intermediate products are also generated in pyrolysis (Tsuchimoto and Kajikawa, 2022). Rubber and plastic produce gases with higher calorific values. Pyrolysis takes moderate to high temperatures (from 450 to 500 °C up to 1200 °C) and operates in an oxygen-free environment (Table 1). Based on the nature of polymers, depolymerization or random fragmentation dominates in pyrolysis (Ragaert et al., 2017).

Gasification turns plastic waste into fuel or combustible gases by using air. This low-cost process yields a gaseous mixture of CO, H₂, CO₂,

Table 1
Selected pyrolysis and gasification technologies in chemical recycling method of plastic waste.

Method	Operational details
Pyrolysis technologies	
Akzo process	Circulate fluidized bed system with subsequent combustion. Input consists of a high percentage of PVC and output consists of HCl, CO, H ₂ , CH ₄ , and in some cases hydrocarbons and fly ash.
NRC process	Only PVC is used as the input. Produces purified CaCl ₂ instead of HCl.
PKA process	Use different kinds of waste including municipal solid waste, industrial and plastic waste, and also contaminated soil. The process occurs at 500–550 °C for 45 min in a rotary kiln and results in de-dusted, homogenized CO/H ₂ -rich fuel gas.
PyroMelt process	Combine prior pyrolysis and slagging combustion. Feed several types of waste including post-consumer waste, and yields an elution resistant such as recyclable slag.
BP polymer cracking	Heat fluidized bed at 500 °C in the air absence environment. HCl resulted from PVC.
Gasification technologies	
WGT process	Different types of pre-treated materials feed into a cylindrical reactor at 700–900 °C. Gas and char discharge after the reaction.
Texaco process	Two parts; (i) liquefaction – the plastic waste is thermally cracked into synthetic heavy oil and some condensable and non-condensable gas fractions. The non-condensable gases are reused in the liquefaction as fuel, (ii) entrained bed gasifier – oil and produced condensed gas are injected into the entrained gasifier. Carry out in the presence of oxygen and steam at temperatures 1200–1500 °C. Produce gases are predominantly of CO and H ₂ with some other inert gases, and a small amount of CH ₄ , CO ₂ , and H ₂ O.
SVZ process	Input feed with lignite and waste oil. Oxygen and steam work as gasification media and yield a gas consisting of hydrogen and CO, liquid hydrocarbons, and effluents. A high amount of resulting gas is used for methanol production and 20 % of gas is used for electricity production.

and CH₄ (Dogu et al., 2021). Although gasification produces syngas as the main yield, it also produces impurities such as NH₃, H₂S, NO_x, alkaline metal, and tars (Ragaert et al., 2017). Gasification contains both exothermic and endothermic processes, but the overall process is endothermic. Hydrogenation/hydrocracking is a chemical reaction between H₂ and plastic waste (Table 1). Accordingly, this is processed via a depolymerization process where agglomerated plastic waste is kept at 350 and 400 °C to effect depolymerization and dechlorination. The main yields are HCl, syncrude, hydrogenated solid residue, and off-gas (Al-Salem et al., 2009).

4.2.2.3. Thermal/energy recycling. Plastic waste burns to produce energy in the form of heat, steam, and electricity (Ragaert et al., 2017). Therefore, incineration results in a considerable volume reduction of plastic material. Thermal recycling can be (i) municipal waste combustion, (ii) grate technology, (iii) fluidized bed combustion, and (iv) two-stage incineration. For example, controlled incineration/mass burning of municipal waste (including plastic waste) can convert post-consumer yields into a useful energy type. About 10 % of household energy requirements in Europe are fulfilled through this method. Consequently, this process reduces fossil fuel consumption (Zia et al., 2007). Grate technology refers to the direct one-stage combustion process of waste. Normal municipal waste including regular plastic can incinerate directly under a relatively low temperature of 850 °C. High calorific value fuel can be produced by adding up to 20 % polyurethane to plastic waste (Al-Salem et al., 2009). Fluidized bed combustion techniques can apply to the combustion of municipal solid waste with a high fraction of plastic from polyurethane (PU), and polyethylene (PE) (Zia et al., 2007; Al-Salem et al., 2009). According to the two-stage incineration, polyurethane foams via combustion under grate incineration. Since this method exhausts gas recirculation, CO and NO emissions are minimal (Zia et al., 2007).

4.2.3. Modification

Different physical and chemical modification methods are also applied in the material recovery process. Plastics can incorporate rigid inorganic particles such as CaCO₃, BaSO₄, and fly ash. Physical modification improves the mechanical properties and temperature resistance of novel materials (Gu et al., 2018; Pan et al., 2020). Functional groups can be introduced to an original molecular chain of materials via cross-linking, grafting, and copolymerization. This method is known as a chemical modification. It also improves properties including high impact resistance, heat resistance, aging resistance, storage stability, and rheological properties (Padhan and Sreeram, 2018). These chemically modified plastics include cross-linking of recycled polyethylene, chlorination modification of waste plastic such as recycled polypropylene (PP), and recycled polyvinyl chloride (Pan et al., 2020).

4.2.4. Value addition

Cracking plastics yields value-added substances such as liquids and gaseous fuels (Pan et al., 2020). Catalytic cracking also results in a higher amount of liquid oil with a mixer of polystyrene, polypropylene, and polyethylene (Miandad et al., 2017). In addition, pyrolysis of plastic waste (composed of aromatic hydrocarbons with a small number of aliphatic hydrocarbons) results in a liquid oil with a high heating value (40–45 MJ/kg) which is more similar to the ordinary diesel (Pan et al., 2020). The pyrolyzed polyethylene, polypropylene, polystyrene, polyvinyl chloride (PVC), polyethylene terephthalate (PET), and mixed plastic waste generate gases under different isothermal conditions. These gases are similar to or even better than natural gas (Honus et al., 2016).

Plastics can also be used as construction materials in road construction (e.g., tar), and the production of concrete and brick. Tar consists of different structures and compositions. A significant amount of tar is produced during the co-gasified or co-pyrolysis of plastic waste. In

addition to road construction, tars from plastic waste can be used in various applications such as in the coating industry, and to treat human and animal diseases (Kurt and Isik, 2012). The quality of tar from plastic waste is higher than traditional tar in road construction due to a higher rainfall tolerance, a higher level of ultraviolet resistance, and a longer lifespan (Huang et al., 2022).

The construction industry focuses on lighter materials, as alternatives to natural aggregates in concrete (Rebeiz and Craft, 1995). Plastic waste can be used as an aggregate substitute for concrete. Production of concrete using plastic waste provides additional benefits such as light-weight, sound insulation/soundproofing applications, and thermal insulation (thermal conductivity of five times less than normal composites). However, replacing concrete with higher quantities of plastic waste reduces the strength of materials due to air content and lower bonding capacity. Therefore, the replacement ratio of plastic materials is about 10 to 15 % (Evode et al., 2021). Similarly, the production of bricks is another value-added product for plastic waste. The incorporation of plastic waste with regular brick production materials results in highly stronger bricks, possessing thermal and sound insulation properties (Kognole et al., 2019). Consequently, it is one of the best options to control the available plastic waste with some economic assets.

4.2.5. Decomposition

Biodegradation is an environmentally friendly treatment method as it does not associate with any hazardous by-products (Moharir and Kumar, 2019). However, the oxo-biodegradation process is essential for non-biodegradable plastic waste before the direct degradation/biodegradation through natural organisms. Oxo-biodegradation is an abiotic reaction that pushes plastic towards biodegradation. Non-biodegradable plastics biodegrade very slowly due to the structure of long polymer chains, and the presence of additives such as antioxidants (Zhang et al., 2021). Accordingly, plastics first react with oxygen to oxidize carbon polymers into functional groups such as carboxylic or hydro-carboxylic acids, esters, aldehydes, and alcohols. This reaction transforms fragmented polymers from hydrophobic to hydrophilic (water absorbance) stage. Therefore, oxo-biodegradation determines the rate of biodegradation in non-biodegradable plastics. In addition, oxo-biodegradation fragmentizes plastic waste through photodegradation, thermal degradation, and mechanochemical degradation (Ammala et al., 2011).

Photodegradation absorbs energy from light to cleave the C-C bond of the polymer chain and generates free radicals via several steps including chain scission, crosslinking, and oxidative reactions. The primary oxidation product of photodegradation is the hydroperoxide group (–CH–OOH), which is thermally and photolytically unstable and decomposes to produce radicals. The appearance of carbonyl groups in a degraded polymer thus hints the oxidation and vulnerability to further degradation (Ammala et al., 2011). Next, thermal degradation absorbs heat to initiate oxidation reactions. In mechanochemical degradation, polymer degradation takes place due to sheer and mechanical forces caused by strong ultrasonic radiation. This method is also associated with chemical reactions induced by radicals and resulting in reduced molecular weight.

After oxo-biodegradation, non-biodegradable plastics can decompose via three biodegradation roots including invertebrates, microorganisms, and treated with composite. For example, certain invertebrates can degrade plastics, such as mealworms (e.g., *Tenebrio obscurus* and *Zophobas atratus* for polystyrene), and lesser waxworms (e.g., *Achroia grisella* for high-density polyethylene). Microorganisms also mineralize plastics into CO₂, H₂O, and biomass through biodeterioration (leading to breaking polymer material into smaller particles), depolymerization (enzymes reaction breaks polymer chain into oligomers, dimers, or monomers), bioassimilation (microbial cells absorb small molecules and generate primary and secondary metabolites), and mineralization (release final products such as CO₂, CH₄, H₂O, and other metabolites or gases). Microorganisms decompose unique substances like polyethylene terephthalate (PET, e.g., *Ideonella sakaiensis*), high- and low-density

polyethylene (HDPE/LDPE, e.g., *Penicillium oxalicum*, *Penicillium chrysogenum*), polyvinyl chloride (PVC, e.g., *P. citreonellolis* and *B. flexus*), polyurethane (PU, e.g., *Aspergillus tubingensis*), and polystyrene (PS, e.g., *Psudomenas spp.* and *Bacillus spp.*) (Singh and Sharma, 2008; Kumar and Raut, 2015; Yang et al., 2015; Zhang et al., 2021). A mixing of mature composite with the weighted amount of plastics, and then treatment with existing microorganisms at a certain temperature (range from 40 to 60 °C) is also one of the decomposition methods. This composting method ultimately turns plastics into CO₂, H₂O, CH₄, biomass, and minerals. The composting method is very popular among biodegradable plastics, but it is also used in non-biodegradable plastic waste to reduce environmental deterioration.

4.3. Drawbacks of major plastic waste disposal methods

Implementation of plastic waste management strategies is valuable in mitigating environmental impacts, health repercussions, and socio-economic challenges stemming from plastic waste. Landfilling, mechanical recycling, chemical recycling, and biodegradation are major non-biodegradable plastic waste recycling methods (Al-Salem et al., 2009; Chandran et al., 2020; Lee and Liew, 2021; Damayanti et al., 2022; Huang et al., 2022). These methods have various benefits and drawbacks (Table 2), such as occupying sustainable and economically significant lands for waste disposal (landfilling), complex and expensive processes of converting one plastic item into another, incomplete conversion of the total quantity, substantial energy consumption to maintain efficient thermal degradation, potential lower durability of the recycled product compared to the original, and skilled labors for an effective recycling process, and potential to release toxic gases like dioxins, furans, mercury, polychlorinated biphenyls and hazardous halogens into the atmosphere (Huang et al., 2022).

To overcome those drawbacks, the combined recycling and degradation processes are proposed as a new plastic waste management approach. According to the American Society for Testing and Materials (ASTM) D5033, recycling is classified into four steps. Primary recycling (ASTM 1) involves the mechanical reprocessing of scrap materials, and

transforming them into products with properties equivalent to the original. Secondary recycling (ASTM 2) entails the mechanical reprocessing of used materials into products with reduced properties. Tertiary recycling (ASTM 3) focuses on the recovery of valuable chemical constituents, such as monomers or additives. Quaternary recycling (ASTM 4) is centered on the recovery of energy from the recycling process (Okan et al., 2018). The process of converting waste into more valuable products or materials is referred to as valorization. The term valorizing also extends to the production of energy sources. Consequently, both the tertiary and quaternary steps align with valorization, as they involve the recovery of valuable chemical constituents or the generation of energy (Kibria et al., 2023). In contrast, primary recycling is associated with the reuse of plastics.

5. Controlling plastic waste

5.1. Replacements for plastic

Suitable plastic waste disposal steps such as collecting, sorting, and directing are difficult tasks. Although recycling has been growing steadily over the last 30 years, about 15 % of plastic waste is recycled from annual waste production (Eze et al., 2021). In addition, it requires a considerable amount of money for disposal and recycling methods that may not be financially feasible for underdeveloped countries. Furthermore, degradation and recycling methods can release harmful byproducts into the environment. Since plastic waste management techniques are not fully satisfied, the world is moving again to traditional/alternative materials for replacing plastics with natural polymers (Table 3).

5.2. Public awareness of plastic pollution

Public awareness is important in plastic pollution. It plays a pivotal role as the ultimate driver in the production, use, and consumption of plastic. To address the plastic crisis, individuals should be well-informed about the current situation. By implementing awareness programs and campaigns with the collaboration of global stakeholders, governments can guide people towards change in plastic consumption patterns, sustainable material use, and disposal practices. On World Environmental Day 2023, the United Nations Environmental Program (UNEP) launched a series of awareness activities with the support of some other organizations, individuals, communities, civil society, private partners, and governments worldwide. The World Environment Day highlights diverse community-driven initiatives globally to combat plastic pollution, such as beach clean-ups in Mumbai, cloth-bag sewing workshops in Ghana, zero-plastic-waste concerts in Atlanta, etc. In addition, World Environment Day trends on Twitter with hashtags #WorldEnvironmentDay and #BeatPlasticPollution reflect widespread participation. Overall, these initiatives underscore the role of individuals and

Table 2
Benefits and drawbacks of the major plastic waste management techniques.

Waste disposal method	Benefits	Drawbacks
Landfilling	Low energy requirement, cost-effective, high volume of waste can remove at once, no skilled labor required, easy to construct and adapted to any place.	Exacerbates land shortage, contaminate waterbodies and soils, enters the food chain, releases methane into the atmosphere, and high carbon emission.
Mechanical recycling	Primary recycling causes theoretical zero waste, a straightforward process, and low-skilled labor requirements.	High energy consumption, recycled materials have low durability, require advanced technologies, need to remove external contaminations before the process, generate some byproducts, not suitable for multi-layered waste, and the quality of the material reduces after several cycles.
Chemical recycling	Produces highly valuable products such as liquid and gaseous fuels, as well as char, products enable to resolve current fossil fuel stocks problems.	High technical requirements, clean inputs, results in toxic byproducts, desorption of other substances remaining within the polymer, poorly managed treatment plants may cause more serious issues, and the resulting solids should be in landfills.
Biodegradation	Environmentally friendly than other methods.	A time-consuming process, the abiotic process may result in toxic products.

Table 3
Representative plant and animal-based polymers as an alternative for synthesis polymers.

Material	Polymer	Common biomass source
Plant-based materials		
Cotton	Cellulose	Cotton plant (<i>Gossypium sp.</i>)
Hemp	Cellulose	Hemp (<i>Cannabis sativa</i>)
Flex/Linen	Cellulose	Flex/Linseed (<i>Linum usitatissimum</i>)
Jute	Cellulose and lignin	<i>Corchorus sp.</i>
Ramie	Cellulose	China grass (<i>Boehmeria nivea</i>)
Animal-based materials		
Mohair	Keratin	Angora goat
Angora wool	Keratin	Angora rabbit
Cashmere wool	Keratin	Cashmere goats
Silk	Fibroin	Silk moth (<i>Bombyx mori</i>)
Alpaca wool	Keratin	Alpaca

communities in driving environmental action and thus encourage governments, financial institutions, and industries to invest in large-scale solutions for overcoming the plastic pollution crisis. Furthermore, the World Wildlife Fund (WWF) has launched a public campaign ‘No Plastic in Nature’, aiming to establish a legally binding global agreement to end plastic pollution. Public communication campaigns seek to inform and inspire individuals to alter their behavior, usually within a specific time frame and without commercial gains. These campaigns predominantly utilize various media forms, including mass media, social media, digital platforms, and online channels. For example, ‘Plastic Free July’ is a main

initiative of the Plastic Free Foundation (an Australia-based non-profit organization) that empowers individuals to reduce their single-use plastic consumption (Heidbreder et al., 2021). Similarly, the Pearl Protectors (Sri Lanka-based environmental conservation organization) conducts public awareness to protect Sri Lankan environment by focusing mainly on marine plastic pollution (Ratnayake, 2023).

5.3. Plastic waste trade

The ‘plastic waste trade’ was introduced as a solution to the

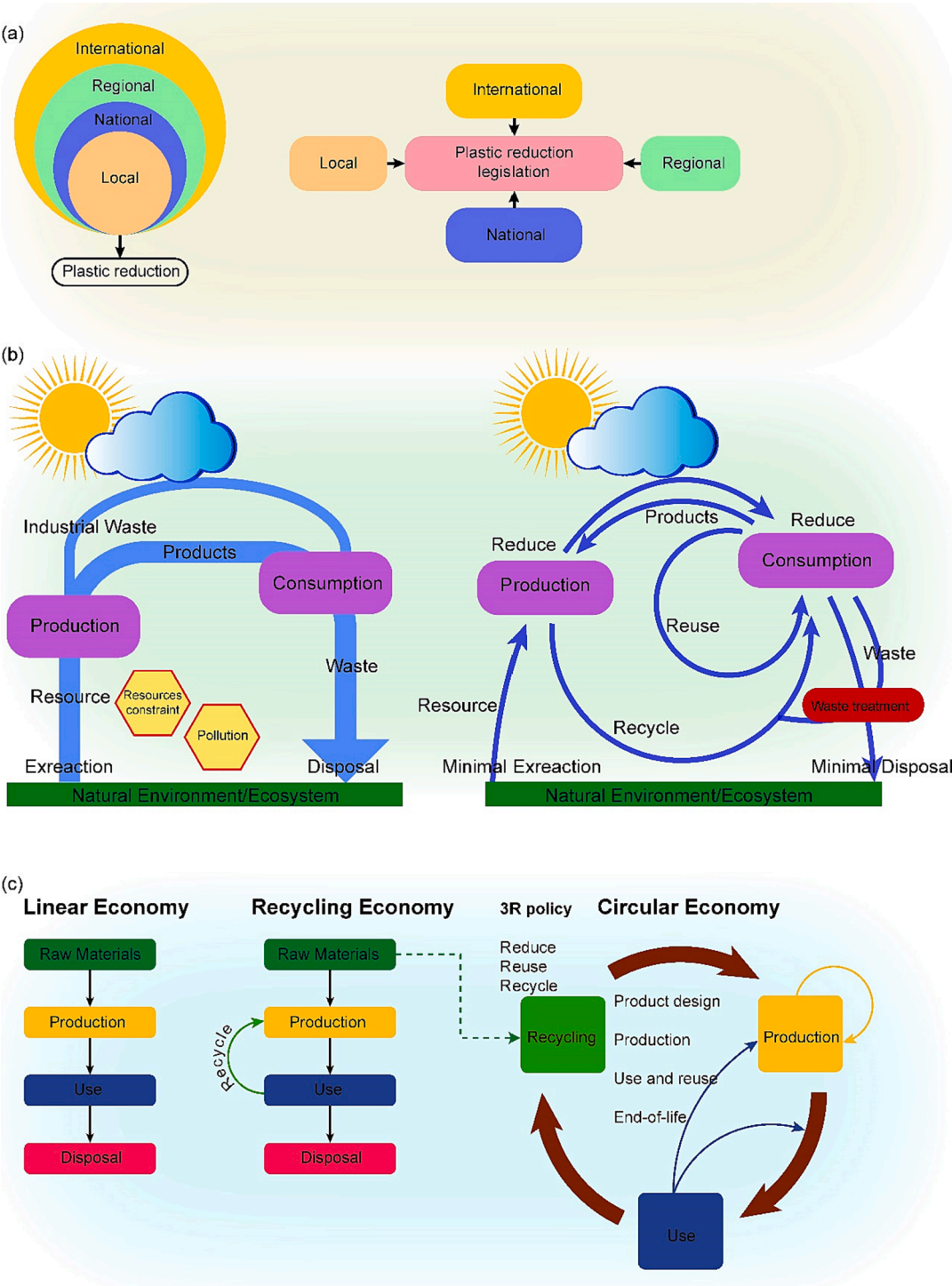


Fig. 7. (a) Legal frameworks for non-effective and effective plastic waste management, (b) production with unsustainable and 3R sustainable resource economy, and (c) potential plastic shifting from a linear economy to a circular economy.

uncontrollable growth of plastic waste in Europe. It is now an international business for the treatment, disposal, and/or recycling of plastic waste (Zhao et al., 2021). Some countries prefer to export their plastic waste due to weaknesses in economically viable post-collection treatment and recycling. The plastic trade can be classified into three stages; before 2000, 2001–2016, and from 2017 until the COVID-19 outbreak (Zhao et al., 2021). The growth rate of global plastic waste was low before 2000, based on the imported and exported volumes. A sharp increase in global plastic waste was observed from 2001 (3.77 million tons) to 2016 (10.99 million tons). However, the global import and export volume trended downward after China announced a ban on plastic waste in 2017 (Zhao et al., 2021).

From the 1990s, plastic demand in China was the driving force of global plastic trade as it accepts more than half of the global plastic waste. China constructed the world's largest plastic waste factory at the beginning of the 21st century. The government of China implemented plastic waste import restrictions, with the implementation of 'green fence' and 'blue sky' programs. Accordingly, the import ban in China generates direct and indirect difficulties for European and North American countries. In addition, the destinations of the plastic waste trade began to change from China to other countries such as Turkey and Indonesia (Wang et al., 2020). However, the total volume of exported plastic waste was reduced in Europe, and inter-Europe trade attempts to manage plastic waste at a stable level within the European countries.

Several facts control plastic waste trade including gate fees and transportation costs, administrative costs, tariff and non-tariff barriers, differences in environmental taxes and policy stringency, differences in treatment capacities, different incentives for recycling/energy recovery, differences in legislation/classification, the requirement of specific technologies, and geographical characteristics of countries/regions. In contrast, the plastic waste trade can cause serious environmental issues in countries with low environmental regulations for recycling and discarding methods (Kellenberg, 2012; Zhao et al., 2021).

5.4. Circular economy to reduce plastic waste

The circular economy is a solution-based principle that includes loops into a material flow through the value chain instead of addressing them in a continuous linear flow (take-make-use-dispose) from production over the use to end-of-life (Fig. 7). It reduces material/resource consumption and retains the resources in the economy by keeping them for a long period (Hahladakis et al., 2020; Syberg et al., 2021, 2022). The UN Sustainable Development Goals also address waste management via circular economic approaches (Schandl et al., 2020). Currently, a substantial portion of plastics follow a linear economy model, contributing to the prevalence of single-use plastics. This usage pattern is the primary cause of ongoing plastic pollution. Therefore, the circular economy is an alternative idea that transforms a linear process into a circular one, emphasizing reduction, reuse, recycling, arrangement, assembly, and circulation (Jang et al., 2020). Accordingly, this concept is built upon three principles: (i) reduce plastic waste and pollution through product design, (ii) retain resources and products in use, and (iii) regenerate and preserve natural systems (Murti et al., 2022). In addition to maintaining materials within the system at the highest value for as long as possible, this concept aims to repair past environmental damage and restore ecological balance (Kunz et al., 2018). The circular economy model (Fig. 7) consists of several attributes such as integrating the operational activities of waste management into the forward supply chain, emphasizing resource recovery through reuse and recycling, lowering emissions, increasing the efficient use of resources, and creating a missing link between circulating material resources and waste management (Babaremu et al., 2022). For example, the circular economy intends to rechannel plastic waste back through useful economic activities. Sometimes, both petrochemical (e.g., polyethylene: PE, polyvinyl chloride: PVC) and bio-based (e.g., polylactic acid: PLA, polyhydroxyalkanoate: PHA) plastics have non-renewable and/or

recyclable circularity potential (Hahladakis et al., 2020). This involves reducing reliance, designing for waste reduction from the beginning, and actively avoiding contamination, toxicity, and pollution (Murti et al., 2022).

The '3R policy' (Reduce, Reuse, Recycle) is a waste management policy on materials recycling (e.g., plastics). Generally, 3R represents reduce (most favored option), reuse, and recycle (least favored option) in waste management (Sakai et al., 2011; Babaremu et al., 2022). Each element in the 3R policy is interrelated (Table 4), and the idea has been built to use resources fully before their disposal (Fig. 7). The 3R policy can be conducted in two ways: (i) on-site promotion (i.e., treatment performed before discharge by the individual at the generation source), and (ii) off-site promotion (i.e., treatment service provided by the government and/or private company after discharge of the waste) (Table 4). To achieve a sustainable circular economy model in plastic waste management, the transition needs to be addressed through a value chain that includes (i) product design, (ii) production, (iii) use and reuse, and (iv) end-of-life.

5.4.1. Product design

The product design includes initiatives facilitating the innovation of more sustainable plastics. One such instance is the EU Directive 2019/904 of June 5, 2019, which targets the reduction of the environmental impact of certain plastics by aiming product requirements (Roy et al., 2022). In addition, producers have a responsibility to encourage and produce plastics that are more easily retained within the circular value chain, thereby reducing environmental losses. Another regulation focused on product design is the EU Commission Regulation 10/2011 of January 14, 2011, on plastic materials, and this article is intended to discuss contact with food. This regulation outlines rules for the composition of food contact materials containing plastics and it permits

Table 4
Different approaches and methods to promote the 3R policy.

Element	Approaches	Detailed methods
Reduce	Governments establish policy and a legal system	Polluter-Pay-Principle Extended producer responsibility Import tax on troublesome goods for solid waste management Promotion of environmentally friendly products Promotion of simplified packaging Green purchasing system (Eco-label system)
	Enhance public consciousness	Buy less and use less Buy environmentally friendly products 'My bag' campaign (carrying reusable bags) Reduction of waste discharge through reuse and recycling at waste generation sources
Reuse	Governments establish policy and legal system	Deposit system for reusable/recyclable products Promotion of returnable containers and packaging Re-use and repair products
	Enhance public consciousness	Buy goods with returnable containers and packaging Sorting reusable goods at generation sources
Recycle	Material recycling	Composting (on-site and off-site) Sorting reusable/recyclable wastes (on-site and off-site)
	Thermal recycling	Refuse Derived Fuel (RDF) production Biomass fuel from chipped green waste Incineration for power generation and heat supply Biogas production

only the use of authorized substances listed in the regulation (Susanti et al., 2020). The European Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) regulates industrial chemicals, including hazardous additives used in plastic products (de Römph and Van Calster, 2018).

5.4.2. Production

Circular economy legislation associated with production aims to ban or levy certain plastic items. The restrictions are more focused on the local level than the global level. Under the Circular Economy concept in China (June 1, 2008), the production of ultra-thin plastic bags was band. Several countries use plastic tax strategies or ban the usage of single plastics (e.g., Germany, Denmark, Ireland, Rwanda, etc.).

5.4.3. Use and reuse

By 2018, 127 countries had implemented regulations addressing plastic bag usage. Ireland implemented taxes in 2002 focusing solely on usage. Accordingly, plastic bag pollution has drastically decreased in Ireland, supported by a drop in total marine litter from 5 % to 0.13 % (Syberg et al., 2021). Similarly, in 2020, the United Kingdom government announced plans to tax a range of single-use plastics by 2022, encouraging a shift towards more sustainable, multiple-use alternatives (Syberg et al., 2021). The road map of Thailand on plastic waste management (2018–2030) aims to reduce and replace some single-use plastics using environmentally friendly products (Samithiwetcharong et al., 2023). Reuse outlines various consumer options such as refilling and reusing containers.

5.4.4. End-of-life

The end-of-life phase brings waste back to production in a new form (Johansen et al., 2022). This phase consists of four subunits: (i) collection and sorting, (ii) recycling (mechanical, chemical, and biological), (iii) life cycle assessment and mass flow analysis of the end-of-life phase, and (iv) policy and regulation (Chandran et al., 2020; Johansen et al., 2022). Life cycle assessment and mass flow analysis of the end-of-life phase play integral roles in comprehensively understanding the environmental impacts and management of plastics within a circular economy. Life cycle assessment evaluates the environmental footprint of a product or system throughout its entire life cycle. The life cycle assessment of recycling technologies reveals that mechanical recycling has generally the least environmental and financial impacts (Johansen et al., 2022; Venkatachalam et al., 2022). Mass flow analysis evaluates the pathways and destinations of plastic waste at the end of its useful life (Kawecki et al., 2018).

5.5. Legislations

The ‘throwaway culture’ of plastic waste causes vast negative consequences, mainly physical and toxicological harm to the ecosystem. This ubiquitous nature of plastic has led policymakers across the globe to develop local and international policies (Table 5), and regulations to reverse and slow down the existing menace (van der Marel, 2022). Legal frameworks related to sustainable materials were initiated nearly four decades ago, and the initial policies focused on material waste reduction (da Costa et al., 2020; Syberg et al., 2021). Lately, several legislations, directives, bans, and levies have been adapted to face growing plastic production at the global, regional, and national levels. Each country adopted different legal systems (policies and legislation) on plastic and plastic waste management to address grievances and consequences presented in society (Table 2 and Fig. 7). Simply, these legislations should promote investments for the transition to a circular economy, elimination of hazardous chemicals from newly designed plastic products, and removal of harmful substances from the environment (Johansen et al., 2022). For example, South Korea introduced ‘a pay-as-you-throw system’ model to control plastic waste at the end of life (Murti et al., 2022). In addition, South Korea adopted several circular

Table 5

Selected international, regional, and country-wise derived policies to control plastic and plastic waste.

Organization	Policy	Function
International organizations on plastic waste		
United Nations (UN)	Sustainable Development Goals (SDG) (2016)	Certify the conservation and sustainable use of all marine resources and it aims to prevent/reduce marine pollution from land activities.
United Nations Environment Programme (UNEP)	Resolution on marine plastic litter and microplastics (MPs) (2019)	Ensure the long-term elimination of MPs and litter in the ocean. Prevent/reduce plastics and MPs from land-based activities.
	Addressing single-use plastic products pollution (2019)	Encourage the promotion of policies to control single-use plastics at national and regional levels, use plastic alternatives, and the improvement of waste management.
	Sustainable consumption and production (2019)	Ensure changes in consumption and production patterns that are reflected in the 2030 agenda of sustainable development.
UNEP, International Union for Conservation of Nature (IUCN), and Life Cycle Initiative	National guidance for plastic pollution hotspotting and shaping.	Guidance is given to the identification of plastic leakages through a framework and tools to assess the progress of the intervention.
World Health Organization (WHO)	Assessment of MPs presence in the environment and impact on human health (2019)	Encourage scrutiny of MPs in the environment and their human health impact. Establish methods to measure MPs and are expected to remove up to 90 %.
United Nations Convention on the Law of the Sea (UNCLOS)	“Constitution for the oceans” 1982, entered into force in (1994)	Constituted an unprecedented attempt at regulating the use of sea resources and forming a stable mankind life (not specified for plastic, but considered as hazardous material such as other marine wastes)
Regional organizations on plastic waste		
European Union (EU)	Reduce the impact of certain plastic products on the environment (2019)	Ensure environmentally sound waste management to prevent and reduce marine litter from both sea and land sources.
Marine Strategy Framework Directive (MSFD)	Achieve Good Environmental Status (GES) of the European marine environment.	Aims to protect the marine environment across Europe while allowing the continuation of sustainable uses of the sea.
Oslo and Paris conservation commission (OSPAR)	Protection of the marine environment of the North-East Atlantic	EU and 15 governments and work towards the protection of the marine environment and its resources. Establish specific guidelines for monitoring marine litter on beaches.
UNEP’s regional seas conventions (RSC)	Protect the coastal and marine environment (1974)	Encompassing 18 regions of the world and promotes the protection of their common marine

(continued on next page)

Table 5 (continued)

Organization	Policy	Function
Regional Action Plan on Marine Litter Management (RAPMaLi)	Aimed to manage marine litter	environment, through a “shared seas” approach. Focused on the education and awareness of the society, encouraging “persons to dispose of waste properly and address the issues of illegal dumping on abandoned beaches and gullies”
Pacific Ocean Pollution Prevention Program (PACOL)	Strategy and work plans on “Clear Pacific”	Comprehensive blueprint details the waste and pollution management priorities of the region by focusing on the different fields.
Association of Southeast Asian Nations (ASEAN)	Combating marine debris in the ASEAN member states (2021–2025)	Committed to reducing plastic release, increasing mop up and reducing leakage, and enhancing waste reuse by value chain creation.
Policy and strategies by countries		
Africa (34 out of 54 countries)	Laws banning plastic bags	The laws impose plastic bag ban/impose levies.
China	Law on the prevention and control of environmental pollution by solid wastes	Regulates waste dumping sites, prohibits plastic dumping in waterbodies, and promotes circular energy.
Korea	Plastic waste control plan	Aimed to reduce plastic waste generation and recycling of generated plastic waste and re-establishment of production and consumption structures and a circular economy.
Malaysia	Road map for zero single-use plastics	Instituted a tax on single-use plastic bags and plastic manufacturers, set up a public awareness unit, and encourages research and development on plastic alternatives.
France	Circular economy law	Banned single-use plastics and promotes circular economic models.
Italy	Plastic packaging law	Imposes a tax on single-use plastics, production, business purchasers, and sellers of plastics.
Sweden	Plastic bag tax	The tax was placed on importers and producers of plastic bags to prevent the spread of MPs and reduce per capita use of plastic bags annually by 2025.
Canada	Canadian environmental protection act (1999)	Addressing plastic pollution using tools at different stages of their life cycle.
USA	Microbeads free water acts (2005)	Prohibited the sale of personal care products containing microbeads and set up a committee to create a response strategy.
Australia	Recycling and waste reduction (2020)	Banned plastic export and provided a flow chart of waste management and recycling.
New Zealand	Waste minimization (microbeads) regulations	Prohibited the sale and manufacture of wash-off

Table 5 (continued)

Organization	Policy	Function
UK	Resource and waste strategy	products containing microbeads. Ensure all plastic packages on the market were recyclable, reusable, or compostable by 2025 and imposed a plastics packaging tax.
Sri Lanka	National environment act	Mainly covers environmental protection, quality, and assessment by prohibiting plastic product use and disposal.

economy strategies, such as the Safe Disposal-Waste Policy in 1980, the Recycling Policy (1992–2003), the Resource Circulation Act in 2016, the Resource Circulation Master Plan (2018–2027), and the Plastic Waste Control Plan (2018–2039, target to reduce plastic waste to 50 % and increase recycle plastic waste to 70 %) (Murti et al., 2022).

5.5.1. Plastic tax

Taxation stands out as an exemplary solution to control plastic pollution. The plastic tax shares conceptual similarities with the carbon tax and is considered a part of green taxation (Desalegn and Tangl, 2022). This taxation aims to reduce plastic consumption by raising the cost of the product through taxes. The revenue generated from these taxes can effectively counterbalance the external costs associated with plastic waste management. In addition, taxes empower consumers by presenting two options: bear the associated tax for plastics or choose alternative options such as paper, without incurring additional costs. This encourages a shift in consumer behavior toward more sustainable choices, fostering environmental awareness (Friedrich, 2021). Many countries (e.g., Italy, Spain, and the United Kingdom) employ taxes to manage plastic pollution, whereas Rwanda selected bans over taxes (Desalegn and Tangl, 2022).

5.5.2. Global collaboration on plastic pollution

Plastic pollution has turned into a global tragedy. Marine plastic pollution is a transboundary impact that occurs due to improper waste management, inadequate social responsibility, littering, illegal dumping, natural disasters, and maritime accidents (Perera et al., 2022; Ratnayake and Perera, 2022). Plastic pollution on a global scale shows challenges in areas beyond national jurisdiction, notably exemplified by the ‘Pacific Garbage Patch’, one of five major accumulation regions for marine plastic debris. Therefore, international cooperation plays a crucial role in tackling the marine plastic crisis beyond local efforts.

The United Nations Convention on the Law of the Sea (UNCLOS) Part XII (articles 192–237) is dedicated to safeguarding and conserving the marine environment. The International Maritime Organization (IMO) plays a crucial role in establishing regulations concerning pollution originating from ships. The Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter 1972 (London Convention) was crafted to ensure effective control over all forms of pollution impacting the marine environment. In addition, the International Convention for the Prevention of Pollution from Ships 1973 (MARPOL) covers the prevention of pollution of the marine environment by ships from operational or accidental causes. The 1978 Protocol to MARPOL, Annex V of MARPOL adopted in 1998 and its revision in 2011 further enhance measures for pollution prevention, especially prohibiting the disposal of plastics anywhere at sea. Furthermore, the United Nations Conference in Environment and Development (UNCED) 1992 Agenda 21 advocates human impacts on the environment. Chapter 17 in UNCED 1992 outlines “protection of the oceans, all kinds of seas, including enclosed and semi-enclosed seas, and coastal areas and the

protection, rational use and development of their living resources". The Conference of the Parties to the Convention of Biodiversity (COP CBD) Scientific and Technical Advisory Panel of the Global Environment Facility released a 2012 report on the impacts of marine debris on marine and coastal biodiversity (GEF 2012). The Global Partnership on Marine Litter (GPML) serves as the coordinating forum for stakeholders at all levels involved in marine debris prevention, and management. Moreover, the European Union (EU) Marine Strategy Framework Directive, the Convention for the Protection of the Marine Environment of the North-East Atlantic (OSPAR), the Baltic Marine Environment Protection Commission, also known as the Helsinki Commission (HELCOM), and the Barcelona Convention address the marine plastic pollution. Establishing such cooperation constitutes significant challenges. The efficacy of these strategies and approaches can be diminished by various factors, including inadequate coordination, the absence of globally binding standards, insufficient national monitoring and reporting, and a lack of comprehensive data on the sources and extent of plastic in the environment (Schandl et al., 2020).

Accordingly, the Global Plastic Pollution Observation system was proposed considering specific objectives including (i) management of analytical methods to quantify and characterize plastic pollution, (ii) collection and making publicly available data on plastic pollution in air, water, soil, and biota, (iii) coordinate global plastic pollution monitoring, and (iv) assess the scale of plastic pollution. These objectives are fulfilled by achieving several tasks such as contributing to global standardization efforts, publishing reporting guidelines, maintaining and curating global data management, and providing knowledge and policy recommendations to necessary parties.

5.5.3. Sustainability and policy linkage

The Sustainable Development Goals (SDGs) also play a prominent role in plastic waste management. Out of 17 goals, 12 goals are connected with plastic management in both direct and indirect ways. For example, SDG 12 (responsible consumption and production: addressing unsustainable plastic practices), SDG 13 (climate action: tackling greenhouse gas emissions in the plastic life cycle), SDG 14 (life below water: focusing on reducing microplastic release into water bodies), and SDG 15 (life on land: mismanagement of plastic waste in terrestrial environments) are directly linked SDGs with plastic waste management (Walker, 2021). Plastic waste management is also indirectly connected to SDG 1 (no poverty: the impact of microplastics on communities), SDG 2 (zero hunger: risks to human health from plastics in food), SDG 3 (good health and well-being: the health impacts of microplastics), SDG 6 (clean water and sanitation: plastics in water sources), SDG 7 (affordable and clean energy: sustainable alternatives to plastic incineration), SDG 9 (industry, innovation, and infrastructure: innovation in sustainable plastics), SDG 10 (reduced inequalities: dealing with the export of plastic waste) and SDG 11 (sustainable cities and communities: tackling the indiscriminate disposal of plastics in areas with inadequate waste management) (de Sousa, 2021; Walker, 2021). These SDGs are required to develop and implement policies that address the environmental, social, and economic aspects of plastic usage. Governments, industries, and international bodies play a pivotal role in shaping policies that promote sustainable plastic management and contribute to achieving the broader SDGs (Kumar et al., 2021).

5.6. Plastic waste management strategies in different countries (Germany and India)

Plastic waste management techniques and strategies are different based on the capacity levels and socio-economic conditions (One et al., 2023). The Plastics Management Index (<https://backtoblueinitiative.com/plastics-management-index/>) measures, compares, and contrasts the actions taken by selected 25 countries to develop plastic waste management strategies, covering the complete lifecycle of plastic products. The objective of this index is to assess the capacity of plastic

management in each country and promote the optimal production and use of plastics as a resource. According to the Plastics Management Index, Germany ranked first, whereas Japan and France ranked second and third, respectively. Therefore, these countries have already implemented policies, regulations, and business practices to deal with current plastic waste management challenges. In contrast, highly populated countries like Brazil (rank = 14), Indonesia (rank = 16), and India (rank = 20) are still in the process of implementing strategies to incorporate consumer actions and perspectives.

Plastic production in Germany was at 21.8 million tons in 2017. Plastic waste (5.1 million tons under the category of packing waste) was used in energy recovery (60.6 %), recycling (38.6 %), and landfilling (0.8 %) (Wurm et al., 2020). Plastic waste covers about 8 % of total waste generated in India, and approximately 6137 tons of plastic waste go uncontrolled per day (Hossain et al., 2022a,b). Per year, around 2.5 million tons of plastic is disposed into landfill sites, more than 1 million tons of waste is sent into incineration, and 0.25 million tons of plastic waste is sent to cement companies as an alternative fuel in cement kilns (Rafey and Siddiqui, 2021).

In Germany, the government has assigned several laws, policies, ordinances, guidelines, and memorandums to establish a well-operated waste management system. In 1991, the German government implemented the German Ordinance of packaging waste and became the first nation to establish legally binding requirements for the recovery and recycling of plastic packaging (Ramasubramanian et al., 2023). Accordingly, Germany became the largest market holder for recycled plastic, and post-consumer plastic waste is higher than pre-consumer plastic waste by a factor of 3.5–5.0 (Ramasubramanian et al., 2023). Germany has established deposit return systems since 2003 to collect plastic packaging waste, and this also led to achieving the highest recycling rate. Germany has reduced plastic waste exports in 2021 (exported only 766,200 tons) which represents a 22.5 % reduction compared to the previous year. This step has increased the plastic waste recycling rate by 70 % (Wurm et al., 2020; Ramasubramanian et al., 2023).

In India, the government has faced difficulties in managing plastic waste, as a lack of space to dispose of the generated waste in highly urbanized lands. India followed mainly primary waste disposal methods such as landfilling/simple disposal to the environment and incineration (Padgelwar et al., 2019). However, the Indian government has recognized integrated plastic waste management strategies such as less use of packing materials, product reuse, use of more durable plastics, and use of green plastic. India has now the capacity to recycle up to 47 % of plastic waste using nearly 20,000 industries with a capacity of 1000 tons per day. This plastic recycling percentage in India is at a very impressive level compared to many developing and developed countries. In addition, the Indian government has launched the most significant cleanliness campaign, 'Swachh Bharat Abhiyan', aimed to create awareness regarding the proper handling of plastic waste and safe disposal. Furthermore, the Ministry of Environment, Forest, and Climate Change has implemented plastic waste management rules to minimize plastic use, segregation, and environmentally friendly recycling, by adopting the polluter pay principle to ensure the sustainability of plastic waste management.

6. Conclusion

Plastic diverts human life to a comfortable path through its vast dynamic properties, such as being lightweight, high strength, design capabilities, good insulation, corrosion resistance, etc. Plastic production has shown rapid growth since the Second World War, and it has become an essential material in modern society. However, plastic waste gains more attention compared to any other type of solid waste due to its high impact on both the environment and humans. The 'throwaway culture', mismanagement of plastic in different stages of its life cycle, and high consumption degrade every ecosystem on Earth. Plastics can

exist in different sizes from macro to nano-scale, and this nature helps to spread it in all kinds of environments, including human bodies. Waste management is a challenging task. Since plastics consist of various characteristics, different waste management practices have been implemented around the world. Biodegradable and non-biodegradable plastic wastes have several end-of-life options such as biological treatments (composting and anaerobic digestion), recycling and reprocessing, incineration with energy recovery, modification reuse, value addition, and landfilling. In addition to disposal methods, several policies, rules, and regulations have been introduced to reduce plastic waste generation and also to manage the existence of waste at global, regional, and national levels. Furthermore, new concepts such as the 3R method, plastic trading, and circular economy promote low material usage and material reuse. Looking at the next few decades, mankind's dependence on plastics is inevitable. Consequently, all the stakeholders should understand the priorities for research, innovation, policies, and legal actions that are required to (i) reduce future plastic usage and (ii) enhance waste management to a better world. Finally, this article emphasized the importance of public awareness and imperative global collaboration in the management of plastic waste.

CRediT authorship contribution statement

P.G.C. Nayanathara Thathsarani Pilapitiya: Writing – review & editing, Writing – original draft, Visualization, Conceptualization.
Amila Sandaruwan Ratnayake: Writing – review & editing, Writing – original draft, Visualization, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We thank W.A.D.B. Weerasingha for assistance in compiling figures. We gratefully acknowledge Jesse Gill, Department of Energy and Mineral Engineering, Pennsylvania State University, USA for his valuable editorial suggestions. The authors would like to acknowledge the financial assistance for this study by the Future Earth Inclusivity and Diversity Participation program.

References

- Adyel, T.M., Macreadie, P.I., 2021. World's largest mangrove forest becoming plastic cesspit. *Front. Mar. Sci.* 8, 766876 <https://doi.org/10.3389/fmars.2021.766876>.
- Alabi, O., Ologbonjaye, K., Awosolu, O., Alalade, O., 2019. Public and environmental health effects of plastic wastes disposal: a review. *J. Toxicol. Risk Assess.* 5, 21. <https://doi.org/10.23937/2572-4061.1510021>.
- Alhazmi, H., Almansour, F.H., Aldhafeeri, Z., 2021. Plastic waste management: a review of existing life cycle assessment studies. *Sustainability* 13, 5340. <https://doi.org/10.3390/su13105340>.
- Al-Mosawi, A.I., Abdulsada, S.A., Rijab, M.A., 2017. Synthesis effect of natural/chemical blend nanoparticles on impact strength of sustainable composite material. *J. Catal. Catal.* 4, 1–5. <https://doi.org/10.37591/jocc.v4i1.2230>.
- Al-Salem, S.M., Lettieri, P., Baeyens, J., 2009. Recycling and recovery routes of plastic solid waste (PSW): a review. *Waste Manag.* 29, 2625–2643. <https://doi.org/10.1016/j.wasman.2009.06.004>.
- Amasuo, E., Baird, J., 2016. The concept of waste and waste management. *J. Manage. Sustain.* 6, 88. <https://doi.org/10.5539/jms.v6n4p88>.
- Ammala, A., Bateman, S., Dean, K., Petinakis, E., Sangwan, P., Wong, S., Yuan, Q., Yu, L., Patrick, C., Leong, K.H., 2011. An overview of degradable and biodegradable polyolefins. *Prog. Polym. Sci.* 36, 1015–1049. <https://doi.org/10.1016/j.progpolymsci.2010.12.002>.
- Andrade, H., Glüge, J., Herzke, D., Ashta, N.M., Nayagar, S.M., Scheringer, M., 2021. Oceanic long-range transport of organic additives present in plastic products: An overview. *Environ. Sci. Eur.* 33, 85. <https://doi.org/10.1186/s12302-021-00522-x>.
- Arabi, S., Nahman, A., 2020. Impacts of marine plastic on ecosystem services and economy: State of South African research. *S. Afr. J. Sci.* 116, 1–7. <https://doi.org/10.17159/sajs.2020/7695>.
- Avery-Gomm, S., Provencher, J.F., Morgan, K.H., Bertram, D.F., 2013. Plastic ingestion in marine-associated bird species from the Eastern North Pacific. *Mar. Pollut. Bull.* 72, 257–259. <https://doi.org/10.1016/j.marpolbul.2013.04.021>.
- Azevedo-Santos, V.M., Brito, M.F.G., Manoel, P.S., Perroca, J.F., Rodrigues-Filho, J.L., Paschoal, L.R.P., Gonçalves, G.R.L., Wolf, M.R., Blettler, M.C.M., Andrade, M.C., Nobile, A.B., Lima, F.P., Ruocco, A.M.C., Silva, C.V., Perbiche-Neves, G., Portinho, J. L., Giarrizzo, T., Arcifa, M.S., Pelicice, F.M., 2021. Plastic pollution: a focus on freshwater biodiversity. *Ambio* 50, 1313–1324. <https://doi.org/10.1007/s13280-020-01496-5>.
- Aznaar, M.P., Caballero, M.A., Sancho, J.A., Francés, E., 2006. Plastic waste elimination by co-gasification with coal and biomass in fluidized bed with air in pilot plant. *Fuel Process. Technol.* 87, 409–420. <https://doi.org/10.1016/j.fuproc.2005.09.006>.
- Babaremu, K.O., Okoya, S.A., Hughes, E., Tijani, B., Teidi, D., Akpan, A., Igwe, J., Karera, S., Oyinlola, M., Akinlabi, E.T., 2022. Sustainable plastic waste management in a circular economy. *Heliyon* 8, E09984. <https://doi.org/10.1016/j.heliyon.2022.e09984>.
- Bhagwat, G., O'Connor, W., Grainge, I., Palanisami, T., 2021. Understanding the fundamental basis for biofilm formation on plastic surfaces: role of conditioning films. *Frontiers in Microbiology* 12, 687118. <https://doi.org/10.3389/fmicb.2021.687118>.
- Boerger, C.M., Lattin, G.L., Moore, S.L., Moore, C.J., 2010. Plastic ingestion by planktivorous fishes in the North Pacific Central Gyre. *Mar. Pollut. Bull.* 60, 2275–2278. <https://doi.org/10.1016/j.marpolbul.2010.08.007>.
- Borrelle, S.B., Ringma, J., Law, K.L., Monnahan, C.C., Lebreton, L., McGivern, A., Murphy, E., Jambeck, J., Leonard, G.H., Hilleary, M.A., Erin, M., Possingham, H.P., De Frond, H., Gerber, L.R., Polidoro, B., Tahir, A., Bernard, M., Mallos, N., Barner, M., Rochman, C.M., 2020. Predicted growth in plastic waste exceeds efforts to mitigate plastic pollution. *Science* 369, 1515–1518. <https://doi.org/10.1126/science.aba3656>.
- Browne, M.A., Niven, S.J., Galloway, T.S., Rowland, S.J., Thompson, R.C., 2013. Microplastic moves pollutants and additives to worms, reducing functions linked to health and biodiversity. *Curr. Biol.* 23, 2388–2392. <https://doi.org/10.1016/j.cub.2013.10.012>.
- Campanale, C., Massarelli, C., Savino, I., Locaputo, V., Uricchio, V.F., 2020. A detailed review study on potential effects of microplastics and additives of concern on human health. *Int. J. Environ. Res. Public Health* 17, 1212. <https://doi.org/10.3390/ijerph17041212>.
- Chandran, M., Tamilkolundu, S., Murugesan, C., 2020. Conversion of plastic waste to fuel. *Plastic Waste Recycl.* 14, 385–399. <https://doi.org/10.1016/B978-0-12-817880-5.00014-1>.
- Chen, R., Li, Q., Xu, X., Zhang, D., 2019. Comparative pyrolysis characteristics of representative commercial thermosetting plastic waste in inert and oxygenous atmosphere. *Fuel* 246, 212–221. <https://doi.org/10.1016/j.fuel.2019.02.129>.
- Chen, H.L., Selvam, S.B., Ting, K.N., Gibbins, C.N., 2021. Microplastic pollution in freshwater systems in Southeast Asia: contamination levels, sources, and ecological impacts. *Environ. Sci. Pollut. Res.* 28, 54222–54237. <https://doi.org/10.1007/s11356-021-15826-x>.
- Chiappone, M., Dienes, H., Swanson, D.W., Miller, S.L., 2005. Impacts of lost fishing gear on coral reef sessile invertebrates in the Florida Keys National Marine Sanctuary. *Biol. Conserv.* 121, 221–230. <https://doi.org/10.1016/j.biocon.2004.04.023>.
- Chouhan, S., Yadav, S.K., Prakash, J., Swati, Singh, S.P., 2014. Effect of Bisphenol A on human health and its degradation by microorganisms: a review. *Ann. Microbiol.* 64, 13–21. <https://doi.org/10.1007/s13213-013-0649-2>.
- Cózar, A., Echevarría, F., González-Gordillo, J.I., Irigoien, X., Úbeda, B., Hernández-León, S., Palma, A.T., Navarro, S., García-de-Lomas, J., Ruiz, A., Fernández-de-Puelles, M.L., Duarte, C.M., 2014. Plastic debris in the open ocean. *Proc. Natl. Acad. Sci. U. S. A.* 111, 10239–10244. <https://doi.org/10.1073/pnas.1314705111>.
- da Costa, J.P., Mouneyrac, C., Costa, M., Duarte, A.C., Rocha-Santos, T., 2020. The role of legislation, regulatory initiatives and guidelines on the control of plastic pollution. *Front. Environ. Sci.* 8, 104. <https://doi.org/10.3389/fenvs.2020.00104>.
- Damayanti, D., Saputri, D.R., Marpaung, D.S.S., Yusupandi, F., Sanjaya, A., Simbolon, Y. M., Asmarani, W., Ulfa, M., Wu, H.-S., 2022. Current prospects for plastic waste treatment. *Polymers* 14, 3133. <https://doi.org/10.3390/polym14153133>.
- Davison, P., Asch, R., 2011. Plastic ingestion by mesopelagic fishes in the North Pacific Subtropical Gyre. *Mar. Ecol. Prog. Ser.* 432, 173–180. <https://doi.org/10.3354/meps09142>.
- de Römph, T.J., Van Calster, G., 2018. Reach in a circular economy: The obstacles for plastics recyclers and regulators. *Rev. Eur. Compar. Int. Environ. Law* 27, 267–277. <https://doi.org/10.1111/reel.12265>.
- de Sousa, F.D., 2021. The role of plastic concerning the sustainable development goals: The literature point of view. *Clean. Respons. Consumpt.* 3, 100020 <https://doi.org/10.1016/j.clrc.2021.100020>.
- Degli, E.M., Morselli, D., Fava, F., Bertin, L., Cavani, F., Viaggi, D., Fabbri, P., 2021. The role of biotechnology in the transition from plastics to bioplastics: An opportunity to reconnect global growth with sustainability. *FEBS Open Bio* 11, 967–983. <https://doi.org/10.1002/2211-5463.13119>.
- Desalegn, G., Tangl, A., 2022. Banning vs taxing, reviewing the potential opportunities and challenges of plastic products. *Sustainability* 14, 7189. <https://doi.org/10.3390/su14127189>.

- Dogu, O., Pelucchi, M., Van de Vijver, R., Van Steenberge, P.H.M., D'hooge, D.R., Cucoi, A., Mehl, M., Frassoldati, A., Faravelli, T., Van Greem, K.M., 2021. The chemistry of chemical recycling of solid plastic waste via pyrolysis and gasification: state-of-the-art, challenges, and future directions. *Prog. Energy Combust. Sci.* 84, 100901 <https://doi.org/10.1016/j.pecs.2020.100901>.
- Egun, N.K., Evbayiro, O.J., 2020. Beat the plastic: an approach to polyethylene terephthalate (PET) bottle waste management in Nigeria. *Waste Dispos. Sustain. Energy* 2, 313–320. <https://doi.org/10.1007/S42768-020-00052-X>.
- Emoyan, O.O., Tesi, G.O., Ohwo, E., Olisah, C., Oghoje, S.U., 2021. Polybrominated diphenyl ethers concentrations in metals and plastics scrap impacted soils: pollution load, sources, ecological, and onsite human health implications. *Environ. Forensic.* <https://doi.org/10.1080/15275922.2021.2006367>.
- Enyoh, C.E., Verla, A.W., Verla, E.N., Ibe, F.C., Amaobi, C.E., 2019. Airborne microplastics: A review study on method for analysis, occurrence, movement and risks. *Environ. Monit. Assess.* 191, 668. <https://doi.org/10.1007/s10661-019-7842-0>.
- Eriksen, M., Lebreton, L.C.M., Carson, H.S., Thiel, M., Moore, C.J., Borerro, J.C., Galgani, F., Ryan, P.G., Reisser, J., 2014. Plastic pollution in the world's oceans: more than 5 trillion plastic pieces weighing over 250,000 tons float at sea. *Public Lib. Sci. One* 9. <https://doi.org/10.1371/journal.pone.0111913>.
- Espinosa, C., Cuesta, A., Esteban, M.A., 2017. Effects of dietary polyvinylchloride microparticles on general health, immune status and expression of several genes related to stress in gilthead seabream (*Sparus aurata* L.). *Fish Shellfish Immunol.* 68, 251–259. <https://doi.org/10.1016/j.fsi.2017.07.006>.
- Espinosa, C., Esteban, M.A., Cuesta, A., 2019. Dietary administration of PVC and PE microplastics produces histological damage, oxidative stress and immunoregulation in European sea bass (*Dicentrarchus labrax* L.). *Fish Shellfish Immunol.* 95, 574–583. <https://doi.org/10.1016/j.fsi.2019.10.072>.
- Evode, N., Qamar, S.A., Bilal, M., Barceló, D., Iqbal, H.M.N., 2021. Plastic waste and its management strategies for environmental sustainability. *Case Stud. Chem. Environ. Eng.* 4, 100142 <https://doi.org/10.1016/j.csee.2021.100142>.
- Eze, W.U., Umunakwe, R., Obasi, H.C., Ugboja, M.I., Uche, C.C., Madufor, I.C., 2021. Plastics waste management: a review of pyrolysis technology. *Clean Technol. Recycl.* 1, 50–69. <https://doi.org/10.3934/ctr.2021.003>.
- Flury, M., Narayan, R., 2021. Biodegradable plastic as an integral part of the solution to plastic waste pollution of the environment. *Curr. Opin. Green Sustainable Chem.* 30, 100490 <https://doi.org/10.1016/j.cogsc.2021.100490>.
- Friedrich, D., 2021. How environmental goals influence consumer willingness-to-pay for a plastic tax: A discrete-choice analytical study. *Environ. Dev. Sustain.* 24, 8218–8245. <https://doi.org/10.1007/s10668-021-01781-7>.
- Gabbott, S., Key, S., Russell, C., Yonan, Y., Zalasiewicz, J., 2020. The geography and geology of plastics. *Plastic Waste Recycl.* 3, 33–63. <https://doi.org/10.1016/B978-0-12-817880-5.00003-7>.
- Gallo, F., Fossi, C., Weber, R., Santillo, D., Sousa, J., Ingram, I., Romano, D., 2018. Marine litter plastics and microplastics and their toxic chemicals components: the need for urgent preventive measures. *Environmental Sciences Europe* 30, 13. In: *Analysis of Nanoplastics and Microplastics in Food* (pp. 159–179). CRC Press. <https://doi.org/10.1186/s12302-018-0139-z>.
- Geyer, R., Jambeck, J.R., Law, K.L., 2017. Production, use, and fate of all plastics ever made. *Sci. Adv.* 3 <https://doi.org/10.1126/sciadv.1700782>.
- Gu, H., Liu, C., Zhu, J., Gu, J., Wujcik, E.K., Shao, L., Wang, N., Wei, H., Scaffaro, R., Zhang, J., Guo, Z., 2018. Introducing advanced composites and hybrid materials. *Adv. Compos. Hybrid Mater.* 1, 1–5. <https://doi.org/10.1007/s42114-017-0017-y>.
- Gunawardhana, G.M.S.S., Perera, U.L.H.P., Ratnayake, A.S., 2023. Sources and fate of plastics into microplastics degradation and remediation methods. In: *Maritime Accidents and Environmental Pollution - the X-Press Pearl Disaster: Causes, Consequences, and Lessons Learned* 8, pp. 155–170. <https://doi.org/10.1201/9781003314301-8>.
- Hahladakis, J.N., Iacovidou, E., Gerassimidou, S., 2020. Plastic waste in a circular economy. *Plastic Waste Recycl.* 19, 481–512. <https://doi.org/10.1016/B978-0-12-817880-5.00019-0>.
- Haider, T.P., Völker, C., Kramm, J., Landfester, K., Wurm, F.R., 2019. Plastics of the future? The impact of biodegradable polymers on the environment and on society. *Angew. Chem. Int. Ed.* 58, 50–62. <https://doi.org/10.1002/anie.201805766>.
- Hassan, T., Srivastava, A.K., Sarkar, S., Majumdar, G., 2022. Characterization of plastics and polymers: A comprehensive study. *IOP Conf. Series: Mater. Sci. Eng.* 1225, 012033 <https://doi.org/10.1088/1757-899x/1225/1/012033>.
- Hauser, R., Calafat, A.M., 2005. Phthalates and human health. *J. Occup. Environ. Med.* 62, 806–818. <https://doi.org/10.1136/oem.2004.017590>.
- Heidbreder, L.M., Lange, M., Reese, G., 2021. #plasticfreejuly – analyzing a worldwide campaign to reduce single-use plastic consumption with Twitter. *Environ. Commun.* 15, 937–953. <https://doi.org/10.1080/17524032.2021.1920447>.
- Honus, S., Kumagai, S., Némček, O., Yoshioka, T., 2016. Replacing conventional fuels in USA, Europe, and UK with plastic pyrolysis gases – part I: experiments and graphical interchangeability methods. *Energ. Convers. Manage.* 126, 1118–1127. <https://doi.org/10.1016/j.enconman.2016.08.055>.
- Hossain, R., Islam, M.T., Ghose, A., Sahajwalla, V., 2022a. Full circle: challenges and prospects for plastic waste management in Australia to achieve circular economy. *J. Clean. Prod.* 368, 133127 <https://doi.org/10.1016/j.jclepro.2022.133127>.
- Hossain, R., Islam, M.T., Shanker, R., Khan, D., Locock, K.E., Ghose, A., Schandl, H., Dhodapkar, R., Sahajwalla, V., 2022b. Plastic Waste Management in India: Challenges, opportunities, and roadmap for Circular Economy. *Sustainability* 14, 4425. <https://doi.org/10.3390/su14084425>.
- Hu, D., Shen, M., Zhang, Y., Li, H., Zeng, G., 2019. Microplastics and nanoplastics: Would they affect global biodiversity change? *Environ. Sci. Pollut. Res.* 26, 19997–20002. <https://doi.org/10.1007/s11356-019-05414-5>.
- Huang, Y., Qing, X., Wang, W., Han, G., Wang, J., 2020. Mini-review on current studies of airborne microplastics: Analytical methods, occurrence, sources, fate and potential risk to human beings. *TrAC Trends Anal. Chem.* 125, 115821 <https://doi.org/10.1016/j.trac.2020.115821>.
- Huang, S., Wang, H., Ahmad, W., Ahmad, A., Vatin, N.I., Mohamed, A.M., Deifalla, A.F., Mehmood, I., 2022. Plastic waste management strategies and their environmental aspects: a scientometric analysis and comprehensive review. *Int. J. Environ. Res. Public Health* 19, 4556. <https://doi.org/10.3390/ijerph19084556>.
- Hurley, R., Horton, A., Lusher, A., Nizzetto, L., 2020. Plastic waste in the terrestrial environment. *Plastic Waste Recycl.* 7, 163–193. <https://doi.org/10.1016/B978-0-12-817880-5.00007-4>.
- Ilyas, M., Ahmad, W., Khan, H., Yousaf, S., Khan, K., Nazir, S., 2018. Plastic waste as a significant threat to environment – a systematic literature review. *Rev. Environ. Health* 33, 383–406. <https://doi.org/10.1515/reveh-2017-0035>.
- Jacob, H., Besson, M., Swarzenski, P.W., Lecchini, D., Metian, M., 2020. Effects of virgin micro- and nanoplastics on fish: Trends, meta-analysis, and perspectives. *Environ. Sci. Tech.* 54, 4733–4745. <https://doi.org/10.1021/acs.est.9b05995>.
- Jambeck, J.R., Geyer, R., Wilcox, C., Siegler, T.R., Perryman, M., Andrady, A., Narayan, R., Law, K.L., 2015. Plastic waste inputs from land into the ocean. *Science* 347, 768–771. <https://doi.org/10.1126/science.1260352>.
- Jang, Y.-C., Lee, G., Kwon, Y., Lim, J., Jeong, J., 2020. Recycling and management practices of plastic packaging waste towards a circular economy in South Korea. *Resour. Conserv. Recycl.* 158, 104798 <https://doi.org/10.1016/j.resconrec.2020.104798>.
- Johansen, M.R., Christensen, T.B., Ramos, T.M., Syberg, K., 2022. A review of the plastic value chain from a circular economy perspective. *J. Environ. Manage.* 302, 113975 <https://doi.org/10.1016/j.jenvman.2021.113975>.
- John, J., Nandhini, A.R., Velayudhaperumal, C.P., Sillanpää, M., 2021. Microplastics in mangroves and Coral Reef Ecosystems: A Review. *Environ. Chem. Lett.* 20, 397–416. <https://doi.org/10.1007/s10311-021-01326-4>.
- Kane, I.A., Clare, M.A., 2019. Dispersion, accumulation, and the ultimate fate of microplastics in deep-marine environments: A review and future directions. *Front. Earth Sci.* 7, 80. <https://doi.org/10.3389/feart.2019.00080>.
- Kane, I.A., Clare, M.A., Miramontes, E., Wogelius, R., Rothwell, J.J., Garreau, P., Pohl, F., 2020. Seafloor microplastic hotspots controlled by deep-sea circulation. *Science* 368, 1140–1145. <https://doi.org/10.1126/science.aba5899>.
- Kang, J.-H., Kondo, F., Katayama, Y., 2006. Human exposure to Bisphenol A. *Toxicology* 226, 79–89. <https://doi.org/10.1016/j.tox.2006.06.009>.
- Katyal, D., Kong, E., Villanueva, J., 2020. Microplastics in the environment: impact on human health and future mitigation strategies. *Environ. Health Rev.* 63, 27–31. <https://doi.org/10.5864/d2020-005>.
- Kawecki, D., Scheeder, P.R., Nowack, B., 2018. Probabilistic material flow analysis of seven commodity plastics in Europe. *Environ. Sci. Tech.* 52, 9874–9888. <https://doi.org/10.1021/acs.est.8b01513>.
- Kedzierski, M., Frère, D., Le Maguer, G., Bruzaud, S., 2020. Why is there plastic packaging in the natural environment? Understanding the roots of our individual plastic waste management behaviours. *Sci. Total Environ.* 740, 139985 <https://doi.org/10.1016/j.scitotenv.2020.139985>.
- Kellenberg, D., 2012. Trading wastes. *J. Environ. Econ. Manag.* 64, 68–87. <https://doi.org/10.1016/j.jeem.2012.02.003>.
- Khan, F., Ahmed, W., Najmi, A., 2019. Understanding consumers' behavior intentions towards dealing with the plastic waste: perspective of a developing country. *Resour. Conserv. Recycl.* 142, 49–58. <https://doi.org/10.1016/j.resconrec.2018.11.020>.
- Khoele, K.K., Gbadeyan, O.J., Chuniell, V., Sithole, B., 2023. The devastation of waste plastic on the environment and remediation processes: A critical review. *Sustainability* 15, 5233. <https://doi.org/10.3390/su15065233>.
- Kibria, M.G., Masuk, N.I., Safayet, R., Nguyen, H.Q., Moursheid, M., 2023. Plastic waste: Challenges and opportunities to mitigate pollution and effective management. *Int. J. Environ. Res.* 17, 20. <https://doi.org/10.1007/s41742-023-00507-z>.
- Kognole, R.S., Shipkule, K., Patil, M., Patil, L., Survase, U., 2019. Utilization of plastic waste for making plastic bricks. *Int. J. Trend Sci. Res. Devel.* 3, 878–880. <https://doi.org/10.31142/ijtsrd23938>.
- Kumar, P., 2018. Role of plastics on human health. *Indian J. Pediatr.* 85, 384–389. <https://doi.org/10.1007/s12098-017-2595-7>.
- Kumar, S., Raut, S., 2015. Microbial degradation of low density polyethylene (LDPE): a review. *J. Environ. Chem. Eng.* 3, 462–473. <https://doi.org/10.1016/j.jece.2015.01.003>.
- Kumar, R., Verma, A., Shome, A., Sinha, R., Sinha, S., Jha, P.K., Kumar, R., Kumar, P., Das, S., Sharma, P., Prasad, P.V., 2021. Impacts of plastic pollution on ecosystem services, sustainable development goals, and need to focus on circular economy and policy interventions. *Sustainability* 13, 9963. <https://doi.org/10.3390/su13179963>.
- Kunz, N., Mayers, K., Van Wassenhove, L.N., 2018. Stakeholder views on extended producer responsibility and the circular economy. *Calif. Manage. Rev.* 60, 45–70. <https://doi.org/10.1177/0008125617752694>.
- Kurt, Y., Isik, K., 2012. Comparison of tar produced by traditional and laboratory methods. *Stud. Ethno-Med.* 6, 77–83. <https://doi.org/10.1080/09735070.2012.11886423>.
- Laist, D.W., 1997. Impacts of marine debris: entanglement of marine life in marine debris including a comprehensive list of species with entanglement and ingestion records. In: *Coe, J.M., Rogers, D.B. (Eds.), Marine Debris. Springer Series on Environmental Management.* Springer, New York, NY. https://doi.org/10.1007/978-1-4613-8486-1_10.
- Lange, J.-P., 2021. Managing plastic waste-sorting, recycling, disposal, and product redesign. *ACS Sustain. Chem. Eng.* 9, 15722–15738. <https://doi.org/10.1021/acsschemeng.1c05013>.

- Law, K.L., Starr, N., Siegler, T.R., Jambeck, J.R., Mallos, N.J., Leonard, G.H., 2020. The United States' contribution of plastic waste to land and ocean. *Sci. Adv.* 6, eabd0288 <https://doi.org/10.1126/sciadv.abd0288>.
- Lebreton, L., Andrady, A., 2019. Future scenarios of global plastic waste generation and disposal. *Palgrave Commun.* 5, 6. <https://doi.org/10.1057/s41599-018-0212-7>.
- Lee, A., Liew, M.S., 2021. Tertiary recycling of plastics waste: an analysis of feedstock, chemical and biological degradation methods. *J. Mater. Cycl. Waste Manage.* 23, 32–43. <https://doi.org/10.1007/s10163-020-01106-2>.
- Liang, Y., Tan, Q., Song, Q., Li, J., 2021. An analysis of the plastic waste trade and management in Asia. *Waste Manag.* 119, 242–253. <https://doi.org/10.1016/j.wasman.2020.09.049>.
- Liu, Q., Schauer, J., 2021. Airborne microplastics from waste as a transmission vector for Covid-19. *Aerosol Air Qual. Res.* 21, 200439 <https://doi.org/10.4209/aaqr.2020.07.0439>.
- Martin, C., Almahasheer, H., Duarte, C.M., 2019. Mangrove forests as traps for marine litter. *Environ. Pollut.* 247, 499–508. <https://doi.org/10.1016/j.envpol.2019.01.067>.
- Masoumi, H., Safavi, S.M., Khani, Z., 2012. Identification and classification of plastic resins using Near Infrared Reflectance Spectroscopy. *Int. J. Mech. Aerosp. Ind. Mechatr. Manuf. Eng.* 6, 877–884.
- McGain, F., McAlister, S., McGavin, A., Story, D., 2010. The financial and environmental costs of reusable and single-use plastic anaesthetic drug trays. *Anaesth. Intensive Care Med.* 38, 538–544. <https://doi.org/10.1177/0310057X1003800320>.
- Mendrik, F.M., Henry, T.B., Burdett, H., Hackney, C.R., Waller, C., Parsons, D.R., Hennige, S.J., 2021. Species-specific impact of microplastics on coral physiology. *Environ. Pollut.* 269, 116238 <https://doi.org/10.1016/j.envpol.2020.116238>.
- Miandad, R., Barakat, M.A., Aburizaiza, A.S., Rehan, M., Ismail, I.M.I., Nizami, A.S., 2017. Effect of plastic waste types on pyrolysis liquid oil. *Int. Biodegrad. Biodegr.* 119, 239–252. <https://doi.org/10.1016/j.ibiod.2016.09.017>.
- Moharir, R.V., Kumar, S., 2019. Challenges associated with plastic waste disposal and allied microbial routes for its effective degradation: a comprehensive review. *J. Clean. Prod.* 208, 65–76. <https://doi.org/10.1016/j.jclepro.2018.10.059>.
- Murti, Z., Dharmawan, S., Soedjati, D., Barkah, A., Rahardjo, P., 2022. Review of the circular economy of plastic waste in various countries and potential applications in Indonesia. *IOP Conf. Series: Earth Environ. Sci.* 1098, 012014 <https://doi.org/10.1088/1755-1315/1098/1/012014>.
- Napper, I.E., Thompson, R.C., 2020. Plastic debris in the Marine Environment: History and future challenges. *Global Chall.* 4, 1900081. <https://doi.org/10.1002/gch2.201900081>.
- Narancic, T., Cerrone, F., Beagan, N., O'Connor, K.E., 2020. Recent advances in bioplastics: Application and biodegradation. *Polymers* 12, 920. <https://doi.org/10.3390/polym12040920>.
- Nikiema, J., Asiedu, Z., 2022. A review of the cost and effectiveness of solutions to address plastic pollution. *Environ. Sci. Pollut. Res.* 29, 24547–24573. <https://doi.org/10.1007/s11356-021-18038-5>.
- Oehlmann, J., Schulte-Oehlmann, U., Kloas, W., Jagnytch, O., Lutz, I., Kusk, K.O., Wollenberger, L., Santos, E.M., Paull, G.C., Van Look, K.J.W., Tyler, C.R., 2009. A critical analysis of the biological impacts of plasticizers on wildlife. *Philos. Trans. R. Soc. London* 364, 2047–2062. <https://doi.org/10.1098/rstb.2008.0242>.
- Okan, M., Aydin, H.M., Barsbay, M., 2018. Current approaches to waste polymer utilization and minimization: A Review. *J. Chem. Technol. Biotechnol.* 94, 8–21. <https://doi.org/10.1002/jctb.5778>.
- Omid, A., Naeemipoor, H., Hosseini, M., 2012. Plastic debris in the digestive tract of sheep and goats: an increasing environmental contamination in Birjand, Iran. *Bull. Environ. Contam. Toxicol.* 88, 691–694. <https://doi.org/10.1007/s00128-012-0587-x>.
- Padgelwar, S., Nandan, A., Mishra, A.K., 2019. Plastic Waste Management and current scenario in India: A Review. *Int. J. Environ. Anal. Chem.* 101, 1894–1906. <https://doi.org/10.1080/03067319.2019.1686496>.
- Padhan, R.K., Sreeram, A., 2018. Enhancement of storage stability and rheological properties of polyethylene (PE) modified asphalt using cross linking and reactive polymer based additives. *Constr. Build. Mater.* 188, 772–780. <https://doi.org/10.1016/j.conbuildmat.2018.08.155>.
- Pan, D., Su, F., Liu, C., Guo, Z., 2020. Research progress for plastic waste management and manufacture of value-added products. *Adv. Compos. Hybrid Mater.* 3, 443–461. <https://doi.org/10.1007/s42114-020-00190-0>.
- Panda, A.K., Singh, R.K., Mishra, D.K., 2010. Thermolysis of waste plastics to liquid fuel: a suitable method for plastic waste management and manufacture of value added products—a world perspective. *Renew. Sustain. Energy Rev.* 14, 233–248. <https://doi.org/10.1016/j.rser.2009.07.005>.
- Pawar, P.R., Shirgaonkar, S.S., Patil, R.B., 2016. Plastic marine debris: sources, distribution and impacts on coastal and ocean biodiversity. *Oceanography* 3, 40–54.
- Perera, U.L.H.P., Subasinghe, H.C.S., Ratnayake, A.S., Weerasingha, W.A.D.B., Wijewardhana, T.D.U., 2022. Maritime pollution in the Indian Ocean after the MV X-Press Pearl accident. *Mar. Pollut. Bull.* 182, 114301 <https://doi.org/10.1016/j.marpolbul.2022.114301>.
- Qi, Y., Beriot, N., Gort, G., Huerta, L.E., Gooren, H., Yang, X., Geissen, V., 2020. Impact of plastic mulch film debris on soil physicochemical and hydrological properties. *Environ. Pollut.* 266, 115097 <https://doi.org/10.1016/j.envpol.2020.115097>.
- Rafey, A., Siddiqui, F.Z., 2021. A review of Plastic Waste Management in India – challenges and opportunities. *Int. J. Environ. Anal. Chem.* 103, 3971–3987. <https://doi.org/10.1080/03067319.2021.1917560>.
- Ragaert, K., Delva, L., Van Geem, K., 2017. Mechanical and chemical recycling of solid plastic waste. *Waste Manag.* 69, 24–58. <https://doi.org/10.1016/j.wasman.2017.07.044>.
- Rahman, M.H., Bhoi, P.R., 2021. An overview of non-biodegradable bioplastics. *J. Clean. Prod.* 294, 126218 <https://doi.org/10.1016/j.jclepro.2021.126218>.
- Ramasubramanian, B., Tan, J., Chellappan, V., Ramakrishna, S., 2023. Recent advances in extended producer responsibility initiatives for plastic waste management in Germany and UK. *Mater. Circul. Econ.* 5, 6. <https://doi.org/10.1007/s42824-023-00076-8>.
- Ratnayake, A.S., 2023. Marine pollution after the MV X-Press Pearl maritime disaster: Societal impacts based on research outcomes. *Soc. Impacts*, 100003. <https://doi.org/10.1016/j.socimp.2023.100003>.
- Ratnayake, A.S., Perera, U.L.H.P., 2022. Coastal zone management in Sri Lanka: a lesson after recent naval accidents. *Mar. Pollut. Bull.* 182, 113994 <https://doi.org/10.1016/j.marpolbul.2022.113994>.
- Rebeiz, K.S., Craft, A.P., 1995. Plastic waste management in construction: technological and institutional issues. *Resour. Conserv. Recycl.* 15, 245–257. [https://doi.org/10.1016/0921-3449\(95\)00034-8](https://doi.org/10.1016/0921-3449(95)00034-8).
- Redondo-Hasselerharm, P.E., Falahudin, D., Peeters, E.T.H.M., Koelmans, A.A., 2018. Microplastic effect thresholds for freshwater benthic macroinvertebrates. *Environ. Sci. Tech.* 52, 2278–2286. <https://doi.org/10.1021/acs.est.7b05367>.
- Rezg, R., El-Fazaa, S., Gharbi, N., Mornagui, B., 2014. Bisphenol A and human chronic diseases: current evidences, possible mechanisms, and future perspectives. *Environ. Int.* 64, 83–90. <https://doi.org/10.1016/j.envint.2013.12.007>.
- Rossi, G., Barnoud, J., Monticelli, L., 2014. Polystyrene nanoparticles perturb lipid membranes. *J. Phys. Chem. Lett.* 5, 241–246. <https://doi.org/10.1021/jz402234c>.
- Roy, A., Naskar, A., Ghosh, A., Adhikari, J., Saha, P., Ghosh, M., 2022. Hybrid plastics and natural materials. *Encycl. Mater.: Plast. Polym.* 3, 288–305. <https://doi.org/10.1016/b978-0-12-820352-1.00075-4>.
- Ruj, B., Pandey, V., Jash, P., Srivastava, V., 2015. Sorting of plastic waste for effective recycling. *Int. J. Appl. Sci. Eng. Res.* 4, 564–571. <https://doi.org/10.6088/ijaser.04058>.
- Saalah, S., Saallah, S., Rajin, M., Yaser, A.Z., 2020. Management of biodegradable plastic waste: A review. *Adv. Waste Process. Technol.* 127–143 https://doi.org/10.1007/978-981-15-4821-5_8.
- Sakai, S.-I., Yoshida, H., Hirai, Y., Asari, M., Takigami, H., Takahashi, S., Tomoda, K., Peeler, M.V., Weichert, J., Schmid-Unterseh, T., Douvan, A.R., Hathaway, R., Hylander, L.D., Fischer, C., Oh, G.J., Jinhui, L., Chi, N.K., 2011. International comparative study of 3R and waste management policy developments. *J. Mater. Cycles Waste Manage.* 13, 86–102. <https://doi.org/10.1007/s10163-011-0009-x>.
- Samithiwetcharong, S., Kullavanijaya, P., Suwantee, K., Chavalparit, O., 2023. Towards sustainability through the circular economy of plastic packaging waste management in Rayong Province, Thailand. *J. Mater. Cycles Waste Manage.* 25, 1824–1840. <https://doi.org/10.1007/s10163-023-01657-0>.
- Schandl, H., Lu, Y., Che, N., Newth, D., West, J., Frank, S., Hatfield-Dodds, S., 2020. Shared socio-economic pathways and their implications for global materials use. *Resour. Conserv. Recycl.* 160, 104866 <https://doi.org/10.1016/j.resconrec.2020.104866>.
- Schnurr, R.E.J., Alboiu, V., Chaudhary, M., Corbett, R.A., Quanz, M.E., Sankar, K., Strain, H.S., Thavarajah, V., Xanthos, D., Walker, T.R., 2018. Reducing marine pollution from single-use plastics (SUPs): A review. *J. Mar. Pollut. Bull.* 137, 157–171. <https://doi.org/10.1016/j.marpolbul.2018.10.001>.
- Shah, M., Rajhans, S., Himanshu, P.A., Archana, U.M., 2021. Bioplastic for future: A review then and now. *World J. Adv. Res. Rev.* 9, 56–67. <https://doi.org/10.30574/wjarr.2021.9.2.0054>.
- Shanker, R., Khan, D., Hossain, R., Islam, M.T., Locock, K., Ghose, A., Sahajwalla, V., Schandl, H., Dhodapkar, R., 2022. Plastic waste recycling: Existing Indian scenario and future opportunities. *Int. J. Environ. Sci. Technol.* 20, 5895–5912. <https://doi.org/10.1007/s13762-022-04079-x>.
- Sigler, M., 2014. The effects of plastic pollution on aquatic wildlife: Current situations and future solutions. *Water Air Soil Pollut.* 225, 2184. <https://doi.org/10.1007/s11270-014-2184-6>.
- Singh, N., Ogunsaitan, O.A., Wong, M.H., Tang, Y., 2022. Sustainable materials alternative to petrochemical plastics pollution: A review analysis. *Sustain. Horizons*. <https://doi.org/10.1016/j.horiz.2022.100016>.
- Singh, B., Sharma, N., 2008. Mechanistic implications of plastic degradation. *Polym. Degrad. Stab.* 93, 561–584. <https://doi.org/10.1016/j.polymdegradstab.2007.11.008>.
- Sokolova, T., Krishna, A., Döring, T., 2023. Paper meets plastic: the perceived environmental friendliness of product packaging. *J. Consum. Res.* <https://doi.org/10.1093/jcr/ucad008>.
- Song, J.H., Murphy, R.J., Narayan, R., Davies, G.B.H., 2009. Biodegradable and compostable alternatives to conventional plastics. *Philos. Trans. R. Soc. London* 364, 2127–2139. <https://doi.org/10.1098/rstb.2008.0289>.
- Stuart, J.N., Watson, M.L., Brown, T.L., Eustice, C., 2001. Plastic netting: an entanglement hazard to snakes and other wildlife. *Soc. Study Amphib. Reptiles* 32, 162–164.
- Stubbins, A., Law, K.L., Muñoz, S.E., Bianchi, T.S., Zhu, L., 2021. Plastics in the earth system. *Science* 373, 51–55. <https://doi.org/10.1126/science.abb0354>.
- Suariga, G., Perold, V., Lee, J.R., Lebourd, F., Aliani, S., Ryan, P.G., 2020. Floating macro- and microplastics around the Southern Ocean: Results from the Antarctic circumnavigation expedition. *Environ. Int.* 136, 105494 <https://doi.org/10.1016/j.envint.2020.105494>.
- Susanti, N.K., Mardiatuti, A., Wardiatno, Y., 2020. Microplastics and the impact of plastic on wildlife: A literature review. *IOP Conf. Series: Earth Environ. Sci.* 528, 012013 <https://doi.org/10.1088/1755-1315/528/1/012013>.
- Sussarellu, R., Suquet, M., Thomas, Y., Lambert, C., Fabioux, C., Pernet, M.E.J., Le Goïc, N., Quillien, V., Mingant, C., Epelboin, Y., Corporeau, C., Guyomarch, J., Robbins, J., Paul-Pont, I., Soudant, P., Huvet, A., 2016. Oyster reproduction is

- affected by exposure to polystyrene microplastics. *Proc. Natl. Acad. Sci. U. S. A.* 113, 2430–2435. <https://doi.org/10.1073/pnas.1519019113>.
- Syberg, K., Nielsen, M.B., Clausen, L.P.W., van Calster, G., van Wezel, A., Rochman, C., Koelmans, A.A., Cronin, R., Pahl, S., Hansen, S.F., 2021. Regulation of plastic from a circular economy perspective. *Curr. Opin. Green Sustain. Chem.* 29, 100462 <https://doi.org/10.1016/j.cogsc.2021.100462>.
- Syberg, K., Nielsen, M.B., Oturai, N.B., Clausen, L.P.W., Ramos, T.M., Hansen, S.F., 2022. Circular economy and reduction of micro(nano)plastics contamination. *J. Hazard. Mater. Adv.* 5, 100044 <https://doi.org/10.1016/j.hazadv.2022.100044>.
- Teuten, E.L., Saquing, J.M., Knappe, D.R.U., Barlaz, M.A., Jonsson, S., Björn, A., Rowland, S.J., Thompson, R.C., Galloway, T.S., Yamashita, R., Ochi, D., Watanuki, Y., Moore, C., Viet, P.H., Tana, T.S., Prudente, M., Boonyatumanond, R., Zakaria, M.P., Akkhang, K., Ogata, Y., Hirai, H., Iwasa, S., Mizukawa, K., Hagino, Y., Imamura, A., Saha, M., Takada, H., 2009. Transport and release of chemicals from plastics to the environment and to wildlife. *Philos. Trans. R. Soc. London* 364, 2027–2045. <https://doi.org/10.1098/rstb.2008.0284>.
- Thushari, G.G.N., Senevirathna, J.D.M., 2020. Plastic pollution in the marine environment. *Heliyon* 6, E04709. <https://doi.org/10.1016/j.heliyon.2020.e04709>.
- Tiwari, R., Azad, N., Dutta, D., Yadav, B.R., Kumar, S., 2023. A critical review and future perspective of plastic waste recycling. *Sci. Total Environ.* 881, 163433 <https://doi.org/10.1016/j.scitotenv.2023.163433>.
- Torres-Agullo, A., Karanasiou, A., Moreno, T., Lacorte, S., 2021. Overview on the occurrence of microplastics in air and implications from the use of face masks during the COVID-19 pandemic. *Sci. Total Environ.* 800, 149555 <https://doi.org/10.1016/j.scitotenv.2021.149555>.
- Townsend, T.G., 2011. Environmental issues and management strategies for waste electronic and electrical equipment. *J. Air Waste Manag. Assoc.* 61, 587–610. <https://doi.org/10.3155/1047-3289.61.6.587>.
- Tsuchimoto, I., Kajikawa, Y., 2022. Recycling of plastic waste: a systematic review using bibliometric Analysis. *Sustainability* 14, 16340. <https://doi.org/10.3390/su142416340>.
- van der Marel, E.R., 2022. Trading plastic waste in a global economy: soundly regulated by the Basel Convention? *J. Environ. Law* 34, 477–497. <https://doi.org/10.1093/jel/eqac017>.
- van Seville, E., Wilcox, C., Lebreton, L., Maximenko, N., Hardesty, B.D., van Franeker, J. A., Eriksen, M., Siegel, D., Galgani, F., Law, K.L., 2015. A global inventory of small floating plastic debris. *Environ. Res. Lett.* 10, 124006 <https://doi.org/10.1088/1748-9326/10/12/124006>.
- Venkatachalam, V., Pohler, M., Spierling, S., Nickel, L., Barner, L., Endres, H., 2022. Design for recycling strategies based on the life cycle assessment and end of life options of plastics in a circular economy. *Macromol. Chem. Phys.* 223, 2200046. <https://doi.org/10.1002/macp.202200046>.
- Verma, R., Vinoda, K.S., Papireddy, M., Gowda, A.N.S., 2016. Toxic pollutants from plastic waste- a review. *Procedia Environ. Sci.* 35, 701–708. <https://doi.org/10.1016/j.proenv.2016.07.069>.
- Vianello, A., Jensen, R.L., Liu, L., Vollertsen, J., 2019. Simulating human exposure to indoor airborne microplastics using a breathing thermal manikin. *Sci. Rep.* 9, 8670 <https://doi.org/10.1038/s41598-019-45054-w>.
- Walker, T.R., 2021. Microplastics and the UN Sustainable Development Goals. *Curr. Opin. Green Sustainable Chem.* 30, 100497 <https://doi.org/10.1016/j.cogsc.2021.100497>.
- Walther, B.A., Bergmann, M., 2022. Plastic pollution of four understudied marine ecosystems: A review of mangroves, seagrass meadows, the Arctic Ocean and the deep seafloor. *Emerg. Top. Life Sci.* 6, 371–387. <https://doi.org/10.1042/etls20220017>.
- Wan, Y., Wu, C., Xue, Q., Hui, X., 2019. Effects of plastic contamination on water evaporation and desiccation cracking in soil. *Sci. Total Environ.* 654, 576–582. <https://doi.org/10.1016/j.scitotenv.2018.11.123>.
- Wang, Y., Huang, J., Zhu, F., Zhou, S., 2021b. Airborne microplastics: A review on the occurrence, migration and risks to humans. *Bull. Environ. Contam. Toxicol.* 107, 657–664. <https://doi.org/10.1007/s00128-021-03180-0>.
- Wang, X., Li, P., Su, M., Zou, X., Duan, L., Zhang, H., 2021a. Characteristics of plastic pollution in the environment: a review. *Bull. Environ. Contam. Toxicol.* 107, 577–584. <https://doi.org/10.1007/s00128-020-02820-1>.
- Wang, Y., Qian, H., 2021. Phthalates and their impacts on human health. *Healthcare* 9, 603. <https://doi.org/10.3390/healthcare9050603>.
- Wang, C., Zhao, L., Lim, M.K., Chen, W.-Q., Sutherland, J.W., 2020. Structure of the global plastic waste trade network and the impact of China's import ban. *Resour. Conserv. Recycl.* 153, 104591 <https://doi.org/10.1016/j.resconrec.2019.104591>.
- Welden, N.A., 2020. The environmental impacts of plastic pollution. *Plastic Waste Recycl.* 8, 195–222. <https://doi.org/10.1016/B978-0-12-817880-5.00008-6>.
- Wichai-utcha, N., Chavalparit, O., 2019. 3Rs Policy and plastic waste management in Thailand. *J. Mater. Cycles Waste Manage.* 21, 10–22. <https://doi.org/10.1007/s10163-018-0781-y>.
- Williams, A.T., Rangel-Buitrago, N., 2019. Marine litter: Solutions for a major environmental problem. *J. Coast. Res.* 35, 648–663. <https://doi.org/10.2112/JCOASTRES-D-18-00096.1>.
- Wojnowska-Baryla, I., Kulikowska, D., Bernat, K., 2021. Effect of bio-based products on Waste Management. *Sustainability* 12, 2088. <https://doi.org/10.3390/su12052088>.
- Wu, J., Eduard, P., Thiyagarajan, S., Noordover, B.A.J., van Es, D.S., Koning, C.E., 2015. Semi-aromatic polyesters based on a carbohydrate-derived rigid diol for engineering plastics. *ChemSusChem* 8, 67–72. <https://doi.org/10.1002/cssc.201402935>.
- Wu, H., Hou, J., Wang, X., 2023. A review of microplastic pollution in aquaculture: Sources, effects, removal strategies and prospects. *Ecotoxicol. Environ. Saf.* 252, 114567 <https://doi.org/10.1016/j.ecoenv.2023.114567>.
- Wurm, F.R., Spierling, S., Endres, H., Barner, L., 2020. Plastics and the environment – current status and challenges in Germany and Australia. *Macromol. Rapid Commun.* 41, 2000351 <https://doi.org/10.1002/marc.202000351>.
- Yang, Y., Yang, J., Wu, W.-M., Zhao, J., Song, Y., Gao, L., Yang, R., Jiang, L., 2015. Biodegradation and mineralization of polystyrene by plastic-eating mealworms: part 1. Chemical and physical characterization and isotopic tests. *Environ. Sci. Tech.* 49, 12080–12086. <https://doi.org/10.1021/acs.est.5b02661>.
- Zhang, W., Sik, O.Y., Bank, M.S., Sonne, C., 2023. Macro- and microplastics as complex threats to coral reef ecosystems. *Environ. Int.* 174, 107914 <https://doi.org/10.1016/j.envint.2023.107914>.
- Zhang, F., Zhao, Y., Wang, D., Yan, M., Zhang, J., Zhang, P., Ding, T., Chen, L., Chen, C., 2021. Current technologies for plastic waste treatment: a review. *J. Clean. Prod.* 282, 124523 <https://doi.org/10.1016/j.jclepro.2020.124523>.
- Zhao, C., Liu, M., Du, H., Gong, Y., 2021. The evolutionary trend and impact of global plastic waste trade network. *Sustainability* 13, 3662. <https://doi.org/10.3390/su13073662>.
- Zia, K.M., Bhatti, H.N., Bhatti, I.A., 2007. Methods for polyurethane and polyurethane composites, recycling and recovery: a review. *React. Funct. Polym.* 67, 675–692. <https://doi.org/10.1016/j.reactfunctpolym.2007.05.004>.